

BONUS QUESTION: HOW DOES FLEXIBLE INCENTIVE PAY AFFECT WAGE RIGIDITY?

Meghana Gaur

John Grigsby

Jonathon Hazell

Abdoulaye Ndiaye*

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Abstract

We introduce dynamic incentive contracts into a model of unemployment dynamics. Our main result is that wage cyclicalities from incentives do not dampen the response of unemployment to shocks. The response of unemployment to a shock is first-order equivalent in economies with flexible incentive pay and economies with exogenously fixed wages. This equivalence is due to effort fluctuations, which make marginal costs rigid even if wages are flexible.

*Gaur: Princeton (meghana.gaur@princeton.edu); Grigsby: Princeton (john.grigsby@princeton.edu); Hazell: LSE (J.Hazell@lse.ac.uk); Ndiaye: NYU Stern (andriaye@stern.nyu.edu). We thank Adrien Auclert, Gadi Barlevy, Matthias Doepke, Martin Eichenbaum, Masao Fukui, Xavier Gabaix, Mark Gertler, Mikhail Golosov, Veronica Guerrieri, Kilian Huber, Kyle Herkenhoff, Greg Kaplan, Andre Kurmann, Chen Lian, Alan Manning, Paolo Martellini, Giuseppe Moscarini, Emi Nakamura, Ricardo Reis, Richard Rogerson, Christopher Sandmann, Rob Shimer, Kjetil Storesletten, Venky Venkateswaran, and numerous seminar participants for helpful comments. This paper was previously titled “Bonus Question: Does Flexible Incentive Pay Dampen Unemployment Dynamics?”

1 Introduction

Macroeconomists have long argued that wage rigidity is important for business cycles. If wages do not respond to shocks, then marginal costs vary little. The consequence is small movements in inflation and larger movements in unemployment over the business cycle (Hall, 2005; Gertler, Sala and Trigari, 2008; Blanchard and Galí, 2010; Christiano, Eichenbaum and Trabandt, 2016).

However, there is a debate about whether wages are rigid in the data. A major source of disagreement relates to incentive pay. Incentive pay, such as bonuses, are common—half of U.S. workers receive some incentive pay, including 30% of bottom-decile earners (Lemieux, Macleod and Parent, 2009; Makridis and Gittleman, 2018). Incentive pay adjusts relatively often: bonuses change regularly for individual workers, but base pay tends to remain fixed (Grigsby, Hurst and Yildirmaz, 2021). Moreover incentive pay appears more cyclical than base pay in some studies, including for newly-hired workers (Bils, 1985; Solon, Whatley and Stevens, 1997; Swanson, 2007; Hazell and Taska, 2022; Bils, Kudlyak and Lins, 2022a). One open question is how incentives affect the response of the economy to shocks.

This paper investigates how incentive pay affects wage rigidity in a model of dynamic incentive contracts. We consider a setting with moral hazard and persistent idiosyncratic shocks similar to Edmans, Gabaix, Sadzik and Sannikov (2012), which we embed in a standard labor search model (Mortensen and Pissarides, 1994). When a risk-neutral firm meets a risk-averse worker, the firm offers a contract that flexibly adjusts pay in response to both aggregate and idiosyncratic shocks. The optimal contract trades off incentivizing effort, which is unobservable, with insuring the worker. In the baseline model, there is no bargaining and workers' utility does not respond to shocks. Therefore, all fluctuations in wages are due to incentives. Wages can be procyclical if firms wish to incentivize workers more during booms.

Our main result is that wage cyclicalities due to incentives does not affect the response of unemployment to shocks. The impulse response of labor market tightness to shocks is the same in both the flexible incentive pay economy and an economy with exogenously fixed real wages (Hall, 2005) so long as both economies must be calibrated to the same steady-state labor share. Therefore procyclical wages from incentives do not reduce the response of unemployment to business cycle shocks: the response of unemployment would be the same if instead wages were fixed.¹

This result may be surprising. A standard argument is that flexible incentive pay dampens the response of unemployment to shocks, by reducing marginal costs during contractions

¹Our result does not hinge on the distinction between new hires' and incumbents wages (Pissarides, 2009). The equivalence holds even if the present value of wages of a newly hired worker is procyclical.

(e.g. Weitzman, 1986). The intuition behind our contrasting result relates to incentives. In labor search models, the response of profits to shocks determines vacancy posting and unemployment. With flexible incentive pay, wages may fall during a contraction. Profits and unemployment respond by less—which is the standard effect of wage flexibility. However in our model there is a less standard *incentive effect*, which acts in the opposite direction. A decline in wages lowers incentives and reduces effort, amplifying the response of profits and unemployment. Under the optimal incentive contract, the standard effect and the incentive effect cancel out exactly, because of an envelope theorem. Therefore, profits in the flexible incentive pay economy behave as if neither wages nor effort had responded to the aggregate shock—that is, as if wages were rigid. Put differently, marginal costs are rigid even if wages are procyclical. This result holds with general preferences and arbitrarily persistent shock processes. Characterizing dynamic incentive contracts outside special cases is difficult (e.g. Golosov, Tsyvinski and Werquin, 2016). However, our argument invokes the optimality of the contract without explicitly characterizing the contract’s form.²

As is well known, wage cyclicalities arising from workers receiving more utility during booms—due to bargaining or better outside options—does dampen the response of unemployment. This form of wage cyclicalities arises in standard models without incentives, which are often calibrated to match the comovement of wages and unemployment (e.g. Pissarides, 2009). If the comovement of wages and unemployment is in fact due to incentives, then standard models will understate the “effective” degree of wage rigidity.

Let us mention two caveats. First, our equivalence result applies to the impulse response of unemployment to business cycle shocks, which is an object of common interest in macroeconomics. For instance, the impulse response of unemployment to shocks determines the slope of the Phillips Curve for price inflation in models with nominal rigidity and search (Christiano et al., 2016). However, the response of other variables will differ between the incentive pay and rigid wage models. Notably, labor productivity and output dynamics will differ between the two economies because of the endogeneity of effort, evoking a notion of capacity utilization (Burnside, Eichenbaum and Rebelo, 1993). Second, our mechanism depends on effort and wages moving in the same direction over the business cycle. This pattern is consistent with the available time series evidence, but effort is hard to measure.³

Related literature. A large literature has developed models consistent with the micro-

²We also establish a similar result with endogenous separations (Mortensen and Pissarides, 1994) and limited worker commitment.

³Various measures of worker effort—from time use surveys, variable capacity utilization, and information on workplace injuries—fall during recessions (Burda, Genadek and Hamermesh, 2020; Fernald, 2014; Galí and Van Rens, 2021). Further, the pass-through of idiosyncratic firm shocks to wages is procyclical (Chan, Salgado and Xu, 2023), consistent with firms seeking to incentivize more effort during booms.

evidence on price setting but tractable enough to allow the study of aggregate rigidity, in part via an analytical equivalence to simpler models (e.g., Alvarez, Le Bihan and Lippi, 2016; Auclert, Rigato, Rognlie and Straub, 2022). In parallel, other papers try to isolate which micro moments on price setting are most relevant for aggregate price rigidity (e.g., Kehoe and Midrigan, 2008; Eichenbaum, Jaimovich and Rebelo, 2011). We aim to provide a model that is consistent with the micro-evidence on *wage* setting, while providing an equivalence to simpler models with rigid wages.

Many papers relate wage dynamics to unemployment fluctuations, and argue that wage rigidity amplifies the response of unemployment to business cycle shocks.⁴ However, most papers in this literature abstract from incentive pay, which seems to be important empirically. We provide a model that links incentive pay to unemployment dynamics.

More related to our setting, Moen and Rosén (2011) and Zhou (2022) consider models with incentive contracts and wage posting. They find numerically that incentives amplify unemployment fluctuations. Fongoni (2020) studies a model in which wages affect effort due to exogenous reference-dependent preferences, and finds that the response of effort to wages amplifies business cycle shocks. We provide analytical results that explain how incentives affect unemployment fluctuations, and connect to simpler models with wage rigidity.⁵

2 A Business Cycle Model with Incentive Pay

Labor market. The labor market follows the standard Diamond-Mortensen-Pissarides (DMP) model. Time is discrete. A large measure of risk-neutral firms match with workers and produce output. A unit mass of workers is either employed or unemployed and searching for a job. Let n_t denote the measure of employed workers at the start of period t , while $u_t \equiv 1 - n_t$ is the measure of unemployed workers looking for jobs. Fluctuations in labor market variables are driven by technology, which follows a first-order Markov process $\{z_t\}_{t=0}^{\infty}$ with lower and upper bounds $\underline{z}$ and $\bar{z}$. Denote the history of this process until t by $z^t = \{z_0, \dots, z_t\}$, and denote the marginal distribution of z^t by $\hat{\pi}_t(z^t|z_0)$.

Firms post vacancies v_t to recruit unemployed workers. The number of matches made in period t is given by a constant-returns-to-scale matching function $m(u_t, v_t)$; labor market conditions are summarized by market tightness $\theta_t = v_t/u_t$, with a job finding rate $\phi(\theta_t) =$

⁴An incomplete list of papers from this vast literature includes Shimer (2005); Hall (2005); Hall and Milgrom (2008); Gertler and Trigari (2009); Elsby (2009); Rudanko (2009); Brügemann and Moscarini (2010); Kennan (2010); Gertler, Huckfeldt and Trigari (2020); Fukui (2020); Blanco, Drenik, Moser and Zaratiegui (2022) and Elsby and Gottfries (2022).

⁵Bils, Chang and Kim (2022b) study a model in which incumbent workers' wages are rigid and effort is contractible. In the model, large employment fluctuations can coexist with flexible new hires' wages. Instead, we study a canonical model of dynamic incentive pay with non-contractible effort.

$m(u_t, v_t)/u_t$ and a vacancy filling rate $q_t \equiv q(\theta_t) = m(u_t, v_t)/v_t$. Let $\nu_t \equiv -d \ln q_t / d \ln \theta_t$ denote the period t elasticity of the job filling rate with respect to θ_t . Maintaining a vacancy has a per-period cost of κ . At the end of period $t - 1$, an exogenous fraction s of workers separate from employment and enter unemployment. The unemployed search for new jobs, so u_t evolves as

$$u_t = u_{t-1} + s(1 - u_{t-1}) - \phi(\theta_{t-1})u_{t-1}(1 - s). \quad (1)$$

Preferences and consumption. Workers have time-separable risk-averse preferences over consumption $c_t \in [\underline{c}, \bar{c}]$ and effort $a_t \in [\underline{a}, \bar{a}]$ and discount future payoffs by a factor $\beta \in (0, 1)$. Preferences are summarized by $u(c, a)$, where u is strictly increasing and strictly concave in c , strictly decreasing and strictly convex in a , and Lipschitz continuous.

Employed workers consume their wages in each period, with newly hired workers producing output and receiving a wage in the period in which they are hired. Unemployed workers exert no effort and are paid constant unemployment benefits b , receiving flow payoff $\xi \equiv u(b, 0)$. Therefore, the value of an unemployed worker at the start of period t is

$$U(z_t) = \phi(\theta_t) \mathcal{E}(z_t) + (1 - \phi(\theta_t)) (\xi + \beta \mathbb{E}[U(z_{t+1}) | z_t]), \quad (2)$$

where $\mathcal{E}(z)$ is the worker's value if she begins employment when aggregate productivity is z .

Firms and vacancy posting. Firms maximize expected profits with discount factor β . Each firm can employ at most a single worker. Consider a firm i that successfully matches with a worker at time 0 and starts producing in the same period. The firm's output in period t is $y_t = f(z_t, \eta_t)$, where f is strictly increasing and continuously differentiable in its arguments and η_t is an idiosyncratic shock to the firm's output that is independently distributed across firms.

At the beginning of the period, before the current value of η_t is realized, the worker exerts effort a_t that affects the distribution of idiosyncratic shocks. We assume a general process for η_t , which allows for arbitrary persistence and depends on the worker's effort. The process has lower and upper bounds $\underline{\eta}$ and $\bar{\eta}$, respectively. Define a history of idiosyncratic shocks $\eta^t = \{\eta_0, \dots, \eta_t\}$. The process for η_t is a probability measure $\pi_t(\eta_t | \eta^{t-1}, a^t)$, which gives the probability of η_t being realized given the history η^{t-1} of past idiosyncratic shocks and the worker's history of actions $a^t = \{a_0, \dots, a_t\}$. Thus, worker effort affects output by shifting the distribution of η realizations.

Let $J(z_0)$ denote the firm's value if it matches with a worker in some initial period $t = 0$

when aggregate productivity is z_0 . The value for a firm of posting a vacancy at time 0 is

$$\Pi_0(z_0) \equiv q(\theta_0)J(z_0) - \kappa. \quad (3)$$

Free entry into vacancy posting guarantees that this value is zero in equilibrium. We entertain two possibilities for wage setting.

Flexible incentive pay economy. In this economy, wages are set according to a dynamic incentive contract. The firm observes the initial value of z_0 and will later observe all realizations of aggregate shocks $\{z_t\}_{t=0}^\infty$. Firms additionally observe idiosyncratic shocks η_t in every period of the match. However, they do not observe worker effort a_t . They thus cannot observe whether an output realization is high because the worker exerted high effort or received a lucky idiosyncratic shock, introducing a moral hazard problem.

When a firm and worker meet, the firm offers the worker a contract to incentivize effort and maximize firm value. A contract specifies a wage function mapping idiosyncratic shocks and aggregate productivity to realized wages. The contract does not condition on the worker's effort, which is unobservable to the firm, but "recommends" an effort level given the history of aggregate and idiosyncratic shocks.

Let $(\mathbf{w}, \mathbf{a})$ denote a contract, with $\mathbf{w} \equiv \{w_t(\eta^t, z^t)\}_{t=0, \eta^t, z^t}^\infty$ and $\mathbf{a} \equiv \{a_t(\eta^{t-1}, z^t)\}_{t=0, \eta^{t-1}, z^t}^\infty$ so that the contract is dynamic and state contingent.⁶ Let $\mathcal{X}$ denote the space of possible contracts.

Value of a filled vacancy. Under the contract $(\mathbf{w}, \mathbf{a})$ and at initial productivity z_0 , the firm's expected present value of profits from a filled vacancy is

$$V(\mathbf{w}, \mathbf{a}; z_0) = \sum_{t=0}^{\infty} (\beta(1-s))^t \int \int (f(z_t, \eta_t) - w_t(\eta^t, z^t)) \tilde{\pi}_t(\eta^t, z^t | z_0, \mathbf{a}) d\eta^t dz^t, \quad (4)$$

where $\tilde{\pi}_t(\eta^t, z^t | \mathbf{a}) \equiv \prod_{\tau=0}^t \pi_\tau(\eta_\tau | \eta^{\tau-1}, a^\tau(\eta^{\tau-1}, z^\tau)) \hat{\pi}_\tau(z^\tau | z_0)$ is the probability of observing a realization of η^t and z^t given the initial z_0 and the contracted effort function $\mathbf{a}$ and $a^\tau(\eta^{\tau-1}, z^\tau)$ is the sequence of effort from periods 0 to τ .

The firm's flow profits are the difference between output and wages. The firm forms an expectation over profit realizations by integrating over the distribution of both aggregate and idiosyncratic shocks, the latter of which depends on effort. The risk-neutral firm discounts period- t profits by the economy-wide discount rate β^t and the probability $(1-s)^t$ that the match survives t periods.

⁶Our notation assumes that the workers chooses effort before η_t is known; this is not crucial to the result.

The contract maximizes the value of a filled vacancy

$$J(z_0) = \max_{\{w_t(\eta^t, z^t), a_t(\eta^{t-1}, z^t)\}_{t=0, \eta^t, z^t \in \mathcal{X}}^\infty} V(\mathbf{w}, \mathbf{a}; z_0) \quad (5)$$

subject to the incentive and participation constraints (IC and PC) described below.

Incentive constraint. The worker chooses effort $\tilde{\mathbf{a}} \equiv \{\tilde{a}_t(\eta^{t-1}, z^t)\}_{t=0, \eta^{t-1}, z^t}^\infty$ to maximize utility under the contract. Therefore, the effort suggested by the firm must be incentive compatible; that is, the recommended effort $\mathbf{a}$ must be what is chosen by the worker given the wage contract that the firm offers her. Specifically,

$$\begin{aligned} [\text{IC}] : \mathbf{a} \in \operatorname{argmax}_{\{\tilde{a}_t(\eta^{t-1}, z^t)\}_{t=0, \eta^t, z^t}^\infty} & \sum_{t=0}^\infty (\beta(1-s))^t \left[\int \int u\left(w_t(\eta^t, z^t), \tilde{a}_t(\eta^{t-1}, z^t)\right) \tilde{\pi}_t(\eta^t, z^t | z_0, \tilde{\mathbf{a}}) d\eta^t dz^t \right. \\ & \left. + \beta s \int U(z_{t+1}) \hat{\pi}_{t+1}(z^{t+1} | z_0) dz^{t+1} \right]. \end{aligned} \quad (6)$$

Equation (6) is the value of an employed worker at time 0; the IC requires that the recommended effort maximizes the worker's value given the wage contract offered by the firm. Their value is the sum of two terms. The first is their value conditional on the match's surviving through period t , which occurs with probability $(1-s)^t$. The realized flow payoff to the worker under the contract is her utility from consuming the wage offered by the contract and providing effort, which depends on realizations of aggregate productivity z^t and idiosyncratic productivity η^t . The worker's expected utility integrates over the distribution of aggregate and idiosyncratic productivity shocks. When making effort choices, the worker trades off the disutility from higher effort with the increased probability of realizing a high output draw and, thus, a high wage. The second term of the worker's value is the value conditional on separation. If the contract separates in period t , the worker receives the value of unemployment at the prevailing aggregate productivity z_t . The match separates in period t with probability $(1-s)^{t-1}s$.

Participation constraint. The contract must also promise the worker a value of at least $\mathcal{E}(z_0)$, the "ex ante utility" promised by the firm to the worker at the start of the contract. The constraint is

$$\begin{aligned} [\text{PC}] : \sum_{t=0}^\infty (\beta(1-s))^t & \left[\int \int u\left(w_t(\eta^t, z^t), a_t(\eta^{t-1}, z^t)\right) \tilde{\pi}_t(\eta^t, z^t | z_0, \mathbf{a}) d\eta^t dz^t \right. \\ & \left. + \beta s \int U(z_{t+1}) \hat{\pi}_{t+1}(z^{t+1} | z_0) dz^{t+1} \right] \geq \mathcal{E}(z_0). \end{aligned} \quad (7)$$

The left-hand side of inequality (7) is the worker's value under the contract: it is the objective

function in equation (6) evaluated at the effort choices suggested by the contract.

We assume that firms make take-it-or-leave-it offers to workers and promise a utility $\mathcal{E} = \mathcal{B} \equiv \xi/(1 - \beta)$. As such, wages respond to shocks only due to incentives. Later, we will discuss how our results are affected when wages change due to bargaining or outside options, meaning workers' utility responds to shocks.

Rigid wage economy. Consider a benchmark economy with exogenously rigid wages and effort following the model of Hall (2005). In this economy, wages and effort take constant values $w_t = \bar{w}$ and $a_t = \bar{a}$ for all firms and all t , regardless of realizations of η^t or z^t . The worker's value of employment is the utility from the match and the continuation value vis-à-vis the possibility that the match may separate, which is given by

$$\mathcal{E}(z_0) = \sum_{t=0}^{\infty} (\beta(1-s))^t \left(u(\bar{w}, \bar{a}) + \int \beta s U(z_{t+1}) \hat{\pi}_t(z^{t+1}|z_0) dz^{t+1} \right). \quad (8)$$

The firm's value of a filled vacancy is exogenous and given by

$$J^{\text{rigid}}(z_0) = \sum_{t=0}^{\infty} (\beta(1-s))^t \int (f(z_t, \eta_t) - \bar{w}) \tilde{\pi}_t(\eta^t, z^t|z_0, \bar{\mathbf{a}}) d\eta^t dz^t. \quad (9)$$

The value of a filled vacancy is the expected present discounted value of production minus the rigid wage, where the expectation is taken over realizations of aggregate and idiosyncratic shocks at a fixed effort $\bar{a}$ in all dates and states.

Equilibrium. Given initial unemployment u_0 and a stochastic process $\{z_t, \eta_t\}_{t=0}^{\infty}$, an equilibrium is a collection of functions $\theta(z)$, $J(z)$, $U(z)$, and $\mathcal{E}(z)$, contracts $(\mathbf{w}, \mathbf{a})(z)$, and unemployment rates u_t such that, for all firms, (i) the tightness $\theta(z)$ satisfies the free-entry condition (3) so $\Pi_t = 0$ for all t , (ii) unemployment u_t evolves according to equation (1), (iii) wage and effort functions $(\mathbf{w}, \mathbf{a})(z)$ solve the firm's problem (5)–(7) in the flexible incentive pay economy or $w_t = \bar{w}$ and $a_t = \bar{a}$ in the rigid wage economy, (iv) the value of unemployment $U(z)$ is given by equation (2), (v) the value of employment $\mathcal{E}(z)$ is given by equation (8) in the rigid wage economy or $\mathcal{B}$ in the flexible incentive pay economy, and (vi) the value of a filled vacancy $J(z)$ or $J^{\text{rigid}}(z)$ is given by equation (5) in the flexible incentive pay economy or equation (9) in the rigid wage economy.

3 Equivalence of Incentive Pay and Rigid Wages

This section explains the equivalence between flexible incentive pay and rigid wages. We first explain the intuition in a static limit of our model, before stating the results in the full dynamic model.

3.1 Intuition in the Static Limit

The static version of our economy is the limit as $\beta \rightarrow 0$ and $s \rightarrow 0$, so that agents are myopic and matches last only one period. For transparency, suppose further that a shifts the mean of the distribution of η and the production function is linear, so that output is $y_t = z_t(a_t + \eta_t)$, with η_t an IID mean zero process. The firm's problem becomes

$$\begin{aligned}
 J^{\text{Incentive}}(z) &\equiv \max_{a(z), w(z, y)} \mathbb{E}_\eta [z(a(z) + \eta) - w(z, y)] & (10) \\
 \text{subject to} \quad &a(z) \in \arg \max_{\tilde{a}(z)} \mathbb{E}_\eta [u(w(z, y), \tilde{a}(z))] & [IC] \\
 &\mathbb{E}_\eta [u(w(z, y), a(z))] \geq \mathcal{B}. & [PC]
 \end{aligned}$$

This notation makes explicit that wages may depend on realizations of both z and y (and thus the idiosyncratic component of output $a + \eta$) but that the firm is uncertain over the realized value of η . Let $a^*(z)$ and $w^*(z, y)$ denote the contracted effort and wage levels as a function of productivity and output realizations.⁷

As usual, this contract implies a tradeoff between incentives and insurance. Absent moral hazard, firms would fully insure workers against wage risk. With moral hazard, firms pass idiosyncratic shocks through to workers' wages to provide incentives. This model allows flexible pay since the firm can freely adjust wages subject to the IC and PC without further restrictions. The firm can freely vary wages with z , potentially leading to procyclical wages or effort.

Role of incentives. We now study the response of labor market tightness to shocks to labor demand z . Totally log-differentiating the free entry condition (3) with respect to aggregate productivity z obtains

$$\frac{d \ln \theta}{d \ln z} = \frac{1}{\nu} \cdot \frac{d \ln J}{d \ln z}. \quad (11)$$

⁷Though the mapping is not exact, one can informally think of a bonus as the component of wages associated with incentives, whereas base pay is the component of wages associated with promised utility. For instance, base pay may be the wage payment under the lowest possible realization of η , whereas bonuses may be wage payments above that lowest level.

That is, the elasticity of market tightness with respect to aggregate productivity z is proportional to the elasticity of expected profits per worker to z , where the constant of proportionality depends on the elasticity of vacancy filling rates with respect to vacancies. Moreover, the employment rate n is determined by the job finding rate $f(\theta)$, whose response is proportional to market tightness. Therefore, to understand the response of employment to aggregate productivity shocks, it is sufficient to study the response of profits per worker.

Differentiating expected profits $J(z) \equiv \mathbb{E}_\eta[z(a + \eta) - w]$ with respect to z implies

$$\frac{dJ(z)}{dz} = \overbrace{\mathbb{E}_\eta[a]}^{\text{direct productivity}} - \overbrace{\mathbb{E}_\eta\left[\frac{dw}{dz}\right]}^{\text{wages}} + z \overbrace{\mathbb{E}_\eta\left[\frac{da}{dz}\right]}^{\text{incentives}}. \quad (12)$$

The first-order response of profits to aggregate productivity has three terms. The first is the direct productivity effect: production rises with productivity, *ceteris paribus*. The second is the standard effect of wages: when productivity rises, wages may also increase, which lowers profits. The third term is a less standard incentive effect: effort may respond to aggregate productivity shocks. The direct productivity and wage effects are present in many models. If wages are procyclical, so dw/dz is large, then profits and employment may respond little to productivity shocks.

However, if effort increases with exogenous productivity, then profits may respond strongly even if expected wages are procyclical. Thus, procyclical incentives might offset the effect of wages on profits, leading to large employment responses despite the procyclicality of wages. Wage cyclical dampens the response of unemployment to productivity shocks only insofar as wages move *more* than effort. Therefore incentives matter for employment dynamics regardless of the specific model of wage and effort setting. We now study how wages and effort move under the optimal contract.

Equivalence between rigid wage and flexible incentive pay. We now derive our main result: to a first-order, the response of employment to shocks is the same in a flexible incentive pay economy as in an appropriately calibrated rigid wage economy, even if incentive pay is procyclical. That is, incentive pay is not relevant wage flexibility for determining the response of employment to shocks.

First, consider the response of profits to z in the rigid wage economy. Both the incentive and wage effects are trivially zero because neither effort nor wages respond to z . Therefore, the response of profits to labor demand shocks is just the direct productivity effect: $dJ^{\text{Rigid}}(z)/dz = \bar{a}$.

Second, consider the flexible incentive pay economy. Differentiating profits in the incen-

tive pay economy (10) and applying the envelope theorem, we see that $dJ^{\text{Incentive}}/dz = a^*(z)$. Only the direct productivity effect remains, exactly as in the rigid wage economy.⁸

This result holds because the wage and incentive effects are equal-sized under the optimal contract so that their effects on profits cancel out, leaving only the direct productivity effect. Although wages and effort may adjust, these fluctuations do not (to a first-order) affect the profit of a firm that is optimally choosing effort and wages. This holds even if wages are strongly procyclical under the optimal contract, so that dw/dz is large.

To gain intuition, suppose that an increase in z leads the firm to encourage higher effort, since a and z are complements in production. All else equal, higher effort raises profits. To encourage the worker to provide higher effort, the firm raises the pass-through of idiosyncratic output into wages. The worker then faces more risk, for which she must be compensated with higher average wages. Thus, wages are procyclical and flexible. All else equal, higher wages would lower expected profits.

The effects of higher effort and higher wages on profits, however, exactly cancel each other out. The reason is that under the optimal incentive contract, the firm is indifferent at the margin between increasing expected wages and increasing worker effort. Expected profits respond to productivity shocks as if neither wages nor effort had changed, just as in the rigid wage economy. The response of profits—and thus market tightness—is the same in the rigid wage and flexible incentive pay economies, as long as both economies are calibrated to have the same direct productivity effect ($\bar{a} = a^*$). One can understand this result as stating that, on the optimal incentive contract, the marginal cost to the firm of producing an additional unit of output is rigid.

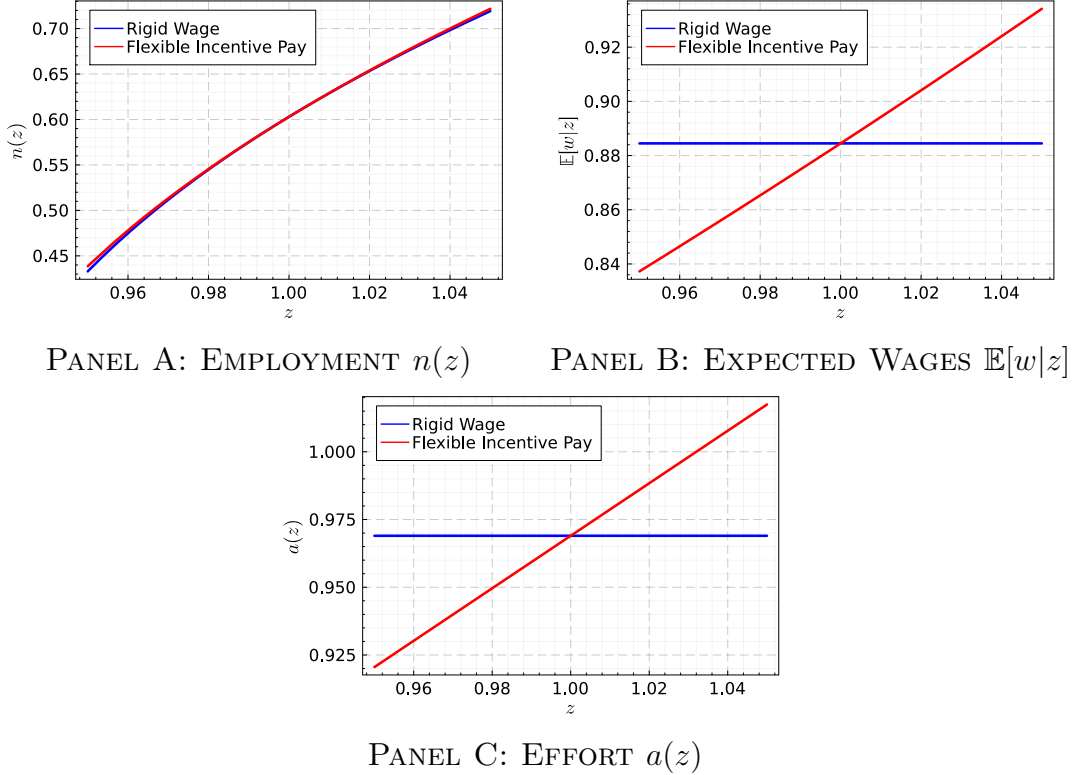
Figure 1 visualizes this result by plotting the behavior of the rigid wage economy (blue line) and the flexible incentive pay economy (red line). Both economies are calibrated to have the same expected wage and effort (and thus profits and employment) when $z = 1$. The horizontal axis of each plot represents exogenous labor productivity z , while the vertical axis plots model-implied employment, expected wages, or effort.

Panel A shows the equivalence of the employment dynamics: the rigid wage and flexible incentive pay economies generate identical responses to aggregate labor productivity z in the neighborhood of $z = 1$. The two models also generate nearly identical employment movements in response to 5% fluctuations in aggregate productivity. This result illustrates the envelope theorem in practice: profit dynamics depend only on the direct productivity effect, which is locally the same in both economies.

Panel B shows that expected wages are procyclical in the incentive pay economy. Therefore, the employment dynamics are the same even though wages fall significantly during

⁸The same result applies if effort is observed and chosen by the firm without an incentive constraint.

Figure 1: Employment, wage and effort fluctuations



Notes: These figures plot the level of employment (Panel A), expected wages (Panel B), and effort (Panel C) as a function of aggregate productivity z in the static model. The red line plots these functions for the flexible incentive pay economy. The blue line plots these functions for the rigid wage economy, calibrated to have the same wage and effort as the flexible incentive pay economy for $z = 1$. We assume that workers have exponential preferences $u(c, a) = -\exp(-r(c - \frac{a^2}{2}))$. The unemployment benefit b is 0.4, η is normally distributed with standard deviation 0.2, and $r = 0.8$. We study the optimal linear contract following Holmstrom and Milgrom (1987).

contractions in the incentive pay economy. Panel C shows the countervailing force: effort also responds in the incentive pay economy. Therefore, incentives offset the stabilizing effect of wages on profits. As a result, large employment responses can coexist with procyclical wages in the incentive pay economy. Wages respond to z in our calibration because a and z are complements. If a and z were not complements, then wages would not be procyclical, but the equivalence result would continue to hold.

Consequences of equivalence. Our result concerns the impulse response of unemployment to shocks. The impulse response is important for several reasons. For instance in labor search models with sticky prices, the impulse response governs the slope of the Phillips Curve for prices (Christiano et al., 2016). As such, the slope of the Phillips Curve for prices can be relatively flat even if wages are procyclical—provided that wage movements are due to incentives and not bargaining. As we have discussed, the reason is that marginal costs do

not vary even if wages do. Moreover with sticky prices, the impulse response of unemployment determines the power of monetary and fiscal policy over the labor market. Therefore unemployment can respond strongly to monetary or fiscal shocks even if wages also respond to the shock.

Our result affects prevailing calibrations of wage rigidity. Researchers typically study labor search models without incentives, in which wages vary only due to bargaining. One often calibrates these models to match estimates of wage cyclicality with respect to unemployment dw/du —see Pissarides (2009) for a classic example. If estimates of dw/du are high, then in standard models, the value promised to the worker falls during a contraction, meaning wages fall in excess of effort (which is commonly assumed to be fixed). As such, marginal costs fall during a contraction and unemployment responds little to shocks.

Our model clarifies that dw/du is not necessarily an informative moment, since the incentive pay model can generate a large unemployment response even if dw/du is large. As we have discussed, dw/dz will be high if the optimal contract requires effort and therefore wages to respond strongly to shocks. However du/dz will be high if, prior to shocks being realized, the direct productivity effect a^* is large. As such, wage cyclicality $dw/du = (dw/dz)/(du/dz)$ can be large even if the impulse response of unemployment du/dz is also large. In order for dw/du to be large, incentive pay must be sufficiently cyclical so that dw/dz is high enough.

Ultimately what matters for calibration is whether wage cyclicality is due to bargaining or incentives. If wages vary due to bargaining in the data, then the standard model and calibration are correct. However, if wages vary due to incentives, standard calibrations matching dw/du will understate the response of unemployment to shocks. Online Appendix A.3 formalizes this discussion in the dynamic version of the model, by allowing wages to vary due to both bargaining and incentives.⁹ One follow-up question is to what extent wage cyclicality in the data is due to bargaining or incentives. Online Appendix B makes one attempt at answering this question in a calibrated version of our dynamic model, and finds that a significant share of wage cyclicality is due to incentives.

Researchers may be able to avoid having to distinguish between bargaining and incentives with other strategies. One could calibrate the degree of wage cyclicality using an estimate of the comovement between wages and output per worker p (e.g. Hagedorn and Manovskii, 2008). This statistic is less affected by incentives, because greater effort raises both the numerator and the denominator. However this strategy is challenging, because empirical work tends to report estimates of dw/du and not dw/dp . One reason is that output per

⁹Formally, we allow workers' utility from the contract $\mathcal{B}(z)$ to fall during contractions, to capture bargaining and changing outside options in a reduced form way. This formulation also evokes a notion of unemployment as a “worker discipline device” (Shapiro and Stiglitz, 1984): if the value of employment is low, workers will offer higher effort at lower wages.

worker has become countercyclical, making estimates of dw/dp hard to interpret (e.g. Galí and Van Rens, 2021).

Two caveats are warranted. First, our equivalence does not apply to the unconditional volatility of unemployment. Unconditional volatility depends both on the impulse response of unemployment and the volatility of the shock. The volatility of labor productivity shocks inferred from the data will differ in models with and without incentives: with incentives, measured labor productivity fluctuations will include both shocks and endogenous movements due to effort.¹⁰ Second, the result does not hold if shocks to z directly perturb the incentive constraints. This might occur if, for instance, the distribution of noise η depends on z .

3.2 Equivalence in the Dynamic Model

We now establish the equivalence of flexible incentive pay and rigid wages result in the dynamic model. This exercise shows that the equivalence result is fairly general. Also, the dynamic model recognizes that employment is a long term contract (Barro, 1977). As such, we can relate the equivalence result to relevant dynamic concepts, such as the present value of wages for new hires.

The main technical challenge for the proof is to transform the problem so that an envelope theorem applies. Common general envelope theorems are not well suited for studying problems with a continuum of nonconvex constraints (e.g. Milgrom and Segal, 2002). The firm's problem has this feature since there is a continuum of incentive compatibility constraints, which are not generally convex. Below, we provide a set of sufficient conditions under which an envelope theorem can be applied to our problem.

Assumption 1. *The set of feasible contracts $(\mathbf{w}, \mathbf{a}) \in \mathcal{X}$ that satisfy the incentive compatibility constraints (6) and participation constraints (7) is nonempty and compact.*

We make the minimal assumption of nonemptiness to allow the optimal contract to exist. We also assume that the set of feasible contracts satisfying the incentive and participation constraints is compact, which allows us to apply a theorem from the applied mathematics literature on sensitivity analysis that applies when there is a continuum of constraints that may not be convex (Bonnans and Shapiro, 2000). Online Appendix A.2 provides two sets of

¹⁰Indeed, the volatility of unemployment will not depend on incentives per se under appropriate calibrations. Equations (11) and (12) imply $Var(d\theta) \propto Var(dJ) = Var(dp) + Var(dw) - 2Cov(dp, dw)$ for $p \equiv \mathbb{E}[z(a + \eta)]$ expected output per worker. This expression for the volatility of tightness does not depend on the model of incentives. Therefore calibrating any model to have the same variance-covariance structure of output per worker p and wages w will generate the same unconditional volatility of market tightness regardless of whether incentives affect wages and output.

sufficient conditions under which the compactness assumption is satisfied. Let $\mathbf{\Gamma}^*(z_0)$ denote the set of optimal contracts $(\mathbf{w}^*, \mathbf{a}^*)$ solving the firm problem (5) given z_0 .

We define an “impulse response” as follows. Denote $z_t = \mathbb{E}[z_t|z_0] + \varepsilon_t$, where, by definition, ε_t is the cumulative innovation to the process for z between 0 and t and ε_0 is known to be 0. We study the response of market tightness to changes in z_0 while holding fixed ε_t for all t , which is the impulse response of market tightness to changes in initial productivity z_0 . The following theorem then holds.

Theorem 1. *Maintain Assumption 1. The first-order impulse response of market tightness to a change in aggregate productivity $d \ln z_0$ is*

$$d \ln \theta_0 = \frac{1}{\nu_0} \frac{\sum_{t=0}^{\infty} (\beta(1-s))^t \frac{\partial}{\partial \ln z_0} \mathbb{E}[f(z_t, \eta_t)|z_0, \mathbf{a}^*] d \ln z_0}{\sum_{t=0}^{\infty} (\beta(1-s))^t \mathbb{E}[f(z_t, \eta_t) - w_t^*|z_0, \mathbf{a}^*]} \quad (13)$$

in the flexible incentive pay economy, for some optimal contract $(\mathbf{w}^, \mathbf{a}^*)$ in $\mathbf{\Gamma}^*(z_0)$, and*

$$d \ln \theta_0 = \frac{1}{\nu_0} \frac{\sum_{t=0}^{\infty} (\beta(1-s))^t \frac{\partial}{\partial \ln z_0} \mathbb{E}[f(z_t, \eta_t)|z_0, \bar{\mathbf{a}}] d \ln z_0}{\sum_{t=0}^{\infty} (\beta(1-s))^t \mathbb{E}[f(z_t, \eta_t) - \bar{w}|z_0, \bar{\mathbf{a}}]}. \quad (14)$$

in a rigid wage economy with $w_t = \bar{w}$ and $a_t = \bar{a}$.

Assume further that (i) the production function f is homogeneous of degree one in aggregate productivity z , (ii) $\partial \mathbb{E}[z_t|z_0]/\partial z_0 = 1$ so that $d \ln z_0$ is a permanent shock, and (iii) the optimal incentive contract at the nonstochastic steady state for z_t is unique. Then, the impulse response of market tightness to z in both economies, in the neighborhood of the nonstochastic steady state for z , is equal to

$$d \ln \theta_0 = \frac{1}{\bar{\nu}} \left(\frac{1}{1 - \Lambda} \right) d \ln z_0, \quad (15)$$

where Λ is the steady-state labor share, defined as

$$\Lambda \equiv \frac{\sum_{t=0}^{\infty} (\beta(1-s))^t \mathbb{E}[w_t|\bar{z}, \mathbf{a}]}{\sum_{t=0}^{\infty} (\beta(1-s))^t \mathbb{E}[f(\bar{z}, \eta_t)|\bar{z}, \mathbf{a}]}, \quad (16)$$

where expectations are evaluated in a steady state with constant aggregate productivity $z_t = \bar{z}$ and $\bar{\nu}$ is the steady-state elasticity of job filling with respect to tightness.

The detailed proof of this theorem appears in Appendix A. There are two key steps in the proof of this theorem. First, we show that the response of firm profits to a shock is the sum of three terms. Just as in the static model, these terms are the direct productivity

effect, and the effect of changes in wages and effort on profits. Second, applying the envelope theorem of Bonnans and Shapiro (2000), the effect of wages and effort on profits sum to zero. Therefore only the direct productivity effect survives—which is the only effect that operates in the rigid wage economy.

Just as in the static model, wage cyclicalities due to incentives does not dampen the response of unemployment to shocks. Different from the static model, what matters in the dynamic model is the present value of wages for new hires. The equivalence holds even if the present value is procyclical. Equation (13) characterizes the impulse response of tightness to labor productivity shocks with flexible incentive pay as the direct productivity effect scaled by the present value of profits; the same is true in the rigid wage economy.¹¹ Therefore, the response of tightness to shocks is the same in both economies, if they feature the same direct productivity effect and the same steady state present value of profits. The final part of the theorem clarifies that flexible incentive pay and rigid wages produce the same tightness response if they are calibrated to the same steady-state labor share. As such, the labor share is a sufficient statistic for the direct productivity effect.¹²

Our result that incentive wage flexibility does not dampen unemployment fluctuations is general. Characterizing the optimal dynamic contract is difficult because of features such as persistent idiosyncratic shocks and potentially nonseparable utility between consumption and effort (Goloso et al., 2016). Applying an envelope theorem allows us to characterize the response of profits without characterizing the optimal contract. Thus, our result holds for general production or utility functions and persistent idiosyncratic shocks.¹³

4 Conclusion

This paper studies how incentive pay affects unemployment dynamics. We embed a dynamic principal-agent problem into a labor search model. Our main result is that wage cyclicalities due to incentives does not dampen the response of unemployment to shocks. The reason is that the effective marginal cost of labor is rigid under the optimal incentive contract, since effort movements offset wage movements. We believe that introducing incentive contracts into macroeconomic models has more applications. For instance, future work could study the effect of incentive contracts on earnings risk, consumption dynamics, or capacity utilization.

¹¹If the optimal contract is not unique, then the impulse response depends on the largest direct productivity effect among optimal contracts when productivity increases and the smallest direct productivity effect among optimal contracts when productivity decreases.

¹²The analytical equivalence is first order but the result holds approximately, in a calibrated and globally solved version of the model (Online Appendix B).

¹³Online Appendix A.4 proves that a version of the result also holds when separations are endogenous because workers have limited commitment.

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ONLINE APPENDIX; NOT FOR PRINT PUBLICATION

A Analytic Appendix

A.1 Proof of Theorem 1

First, we derive the relationship between the impulse response of tightness to TFP shocks and the impulse response of firm value to TFP shocks, which will hold in both the flexible incentive pay and the rigid wage economy. From equation (3), the free entry condition is

$$\begin{aligned} q(\theta_0)J(z_0) - \kappa &= 0 \\ \implies J(z_0) &= \frac{\kappa}{q(\theta_0)} \\ \implies \frac{d \ln \theta_0}{d \ln z_0} &= \frac{1}{\nu_0} \frac{d \ln J(z_0)}{d \ln z_0}. \end{aligned} \tag{17}$$

where $-\nu_0$ is the elasticity of the vacancy filling rate given z_0 . That is, the response of market tightness to aggregate productivity shocks is proportional to the response of the value of a filled job, as in the static model.

Now, we derive the dynamics of firm value and tightness in the rigid wage economy, which will also be a warm-up for deriving the dynamics of tightness in the flexible incentive pay economy. Using equation (9) from the main text, the value of a job in the rigid wage economy is

$$\begin{aligned} J^{\text{rigid}}(z_0) &= \sum_{t=0}^{\infty} (\beta(1-s))^t \mathbb{E}[f(z_t, \eta_t) - \bar{w} | z_0, \bar{\mathbf{a}}] \\ \implies \frac{dJ^{\text{rigid}}(z_0)}{dz_0} &= \sum_{t=0}^{\infty} (\beta(1-s))^t \frac{\partial}{\partial z_0} \mathbb{E}[f(z_t, \eta_t) | z_0, \bar{\mathbf{a}}]. \end{aligned} \tag{18}$$

Using equations (17) and (18) from the Appendix and equation (9) from the main text, tightness dynamics in the rigid wage economy are then

$$\begin{aligned}
\frac{d \ln \theta_0}{d \ln z_0} &= \frac{1}{\nu_0} \frac{d \ln J^{\text{rigid}}(z_0)}{d \ln z_0} = \frac{1}{\nu_0} \frac{z_0}{J^{\text{rigid}}(z_0)} \frac{d J^{\text{rigid}}(z_0)}{d z_0} \\
&= \frac{1}{\nu_0} \frac{z_0 \sum_{t=0}^{\infty} (\beta(1-s))^t \frac{\partial}{\partial z_0} \mathbb{E}[f(z_t, \eta_t) | z_0, \bar{\mathbf{a}}]}{\sum_{t=0}^{\infty} (\beta(1-s))^t \mathbb{E}[(f(z_t, \eta_t) - \bar{w}) | z_0, \bar{\mathbf{a}}]} \\
&= \frac{1}{\nu_0} \frac{\sum_{t=0}^{\infty} (\beta(1-s))^t \frac{\partial}{\partial \ln z_0} \mathbb{E}[f(z_t, \eta_t) | z_0, \bar{\mathbf{a}}]}{\sum_{t=0}^{\infty} (\beta(1-s))^t \mathbb{E}[(f(z_t, \eta_t) - \bar{w}) | z_0, \bar{\mathbf{a}}]}
\end{aligned}$$

which implies the first-order response of log tightness to a change $d \ln z_0$ is

$$d \ln \theta_0 = \frac{1}{\nu_0} \frac{\sum_{t=0}^{\infty} (\beta(1-s))^t \frac{\partial}{\partial \ln z_0} \mathbb{E}[f(z_t, \eta_t) | z_0, \bar{\mathbf{a}}] d \ln z_0}{\sum_{t=0}^{\infty} (\beta(1-s))^t \mathbb{E}[(f(z_t, \eta_t) - \bar{w}) | z_0, \bar{\mathbf{a}}]}, \quad (19)$$

i.e., equation (14) from the main text. Therefore, we have derived the dynamics of tightness in the rigid wage economy.

Next, we turn to dynamics in the flexible incentive pay economy. To start, we must rewrite the firm's problem in the case of flexible incentive pay using the impulse response notation introduced in the main text. Specifically, we let the contracts be given by $(\mathbf{w}, \mathbf{a}) = \{w_t(\eta^t, \varepsilon^t; z_0), a_t(\eta^{t-1}, \varepsilon^t; z_0)\}_{t=0, \eta^t, \varepsilon^t}^{\infty}$ where $w_t(\eta^t, \varepsilon^t; z_0), a_t(\eta^{t-1}, \varepsilon^t; z_0)$ are continuous functions mapping from the history of idiosyncratic and aggregate shocks, and the initial state, to wages and effort. That is, contracts can depend on z_0 and a cumulative set of deviations from z_0 . We use the fact that we consider impulse responses holding fixed a path of deviations to define the measure

$$\pi_t(\eta^t, \varepsilon^t | \mathbf{a}(z_0)) = \prod_{\tau=0}^t \pi_{\tau}(\eta_{\tau} | \eta^{\tau-1}, a^{\tau}(\eta^{\tau-1}, \varepsilon^{\tau}; z_0), \varepsilon^{\tau}) \pi_{\tau}(\varepsilon^{\tau}),$$

where the probability measure does not depend directly on z_0 because η^t is independent of z_0 by assumption, and ε^t does not depend on z_0 by our definition of an impulse response.

Thus, the firm's problem becomes

$$J(z_0) = \max_{\mathbf{w}(z_0), \mathbf{a}(z_0)} \sum_{t=0}^{\infty} (\beta(1-s))^t \int \int (f(\mathbb{E}[z_t | z_0] + \varepsilon_t, \eta_t) - w_t(\eta^t, \varepsilon^t; z_0)) \tilde{\pi}_t(\eta^t, \varepsilon^t | \mathbf{a}(z_0)) d\eta^t d\varepsilon^t \quad (20)$$

subject to participation constraints

$$\sum_{t=0}^{\infty} (\beta(1-s))^t \left[\int \int u \left(w_t(\eta^t, \varepsilon^t; z_0), a_t(\eta^{t-1}, \varepsilon^t; z_0) \right) \tilde{\pi}_t(\eta^t, \varepsilon^t | \mathbf{a}(z_0)) d\eta^t d\varepsilon^t \right. \\ \left. + \beta s \int U(z_{t+1}) \hat{\pi}_t(z^{t+1} | z_0) dz^{t+1} \right] \geq \mathcal{E}(z_0) \quad (21)$$

and incentive compatibility constraints

$$\sum_{t=0}^{\infty} (\beta(1-s))^t \left[\int \int u \left(w_t(\eta^t, \varepsilon^t; z_0), \tilde{a}_t(\eta^{t-1}, \varepsilon^t) \right) \tilde{\pi}_t(\eta^t, \varepsilon^t | \tilde{\mathbf{a}}) d\eta^t d\varepsilon^t \right] \\ \leq \sum_{t=0}^{\infty} (\beta(1-s))^t \left[\int \int u \left(w_t(\eta^t, \varepsilon^t; z_0), a_t(\eta^{t-1}, \varepsilon^t; z_0) \right) \tilde{\pi}_t(\eta^t, \varepsilon^t | \mathbf{a}(z_0)) d\eta^t d\varepsilon^t \right] \quad (22)$$

for all $\tilde{\mathbf{a}} \in \mathcal{X}$. To study profits, we combine the IC and PC into a functional $G(\mathbf{w}, \mathbf{a})$, defined such that $G(\mathbf{w}, \mathbf{a}) \leq 0$ holds if and only if $(\mathbf{w}, \mathbf{a})$ is a feasible contract in $\mathcal{X}$ that satisfies the PC (21) and IC (22). Let $\lambda(z_0)$ denote the costate functional on these constraints, and let $\Phi \equiv \{(\mathbf{w}, \mathbf{a}) \in \mathcal{X} : G(\mathbf{w}, \mathbf{a}) \leq 0\}$ be the set of feasible contracts that satisfy the IC and PC constraints. We write the value of a filled job using the functional Kuhn–Tucker Lagrangian:¹⁴

$$J(z_0) = V(\mathbf{w}^*, \mathbf{a}^*; z_0) - \langle G(\mathbf{w}^*, \mathbf{a}^*; z_0), \lambda^* \rangle, \quad (23)$$

where the star superscripts indicate values under the optimal contract at z_0 . Then, we can decompose the response of firm profits to z_0 , generalizing decomposition (10) from Section 3.1. The response of profits to aggregate shocks in the flexible incentive pay economy is

$$\frac{dJ(z_0)}{dz_0} = \underbrace{\frac{\partial}{\partial z_0} V(\mathbf{w}^*, \mathbf{a}^*; z_0)}_{\text{(A) direct productivity effect on profits}} - \underbrace{\left\langle \frac{\partial}{\partial z_0} G(\mathbf{w}^*, \mathbf{a}^*; z_0), \lambda^*(z_0) \right\rangle}_{\text{(B) direct effect on participation and incentives}} \quad (24) \\ + \underbrace{\sum_{x \in \{\mathbf{w}^*, \mathbf{a}^*\}} [\partial_x V(\mathbf{w}^*, \mathbf{a}^*; z_0) - \langle \partial_x G(\mathbf{w}^*, \mathbf{a}^*; z_0), \lambda^*(z_0) \rangle] \cdot \frac{dx}{dz_0} - \left\langle G(\mathbf{w}^*, \mathbf{a}^*; z_0), \frac{d\lambda^*(z_0)}{dz_0} \right\rangle}_{\text{(C) indirect effects on optimal contract and costates}},$$

where ∂_x represents the vector of partial derivatives with respect to some variable x . The direct productivity effect (A) measures how shocks to initial productivity affect the expected present value of output in all periods, where the expectation conditions on initial productivity

¹⁴The notation $\langle x, x^* \rangle$ denotes the value of the linear functional x^* at a point x . This notation is necessary because there is a continuum of constraints—see Section 3.1.1 of Golosov et al. (2016) for a formal definition of Lagrangians with this notation.

z_0 and contracted effort $\mathbf{a}^*$.

Term (B) captures the effects on the constraints. Since z_0 affects the constraints only indirectly through the contract $(\mathbf{w}, \mathbf{a})$ in our baseline model, there is no direct effect of z_0 on constraints.

The (C) term captures the effects that the shock has on profits through changes in the firm's choice variables. (C) has three pieces. First, the shock may shift the optimal contract's wage function $\mathbf{w}^*$. This is the wage effect: the wage paid for each future realization of η^t and z^t may differ for contracts signed at different initial aggregate productivity levels z_0 . Second, the shock may shift the optimal contract's recommended effort function $\mathbf{a}^*$, which affects output. This is the incentive effect. Finally, the shock may shift the value of the costates on the participation and incentive constraints.

To derive $d \ln J(z_0)/d \ln z_0$ in the flexible incentive pay economy, we seek to apply an envelope theorem. However, it is not trivial to show that an envelope theorem applies in our setting because the firm faces a continuum of constraints that may be non-convex. We, therefore, pursue two proof strategies that rely on different conditions. Our first proof in Section A.1.1 relies on the compactness of the set of incentive compatible mechanisms that satisfy the PC, as assumed in Assumption 1. We provide two alternative sets of conditions guaranteeing this compactness in Section A.2 below: (i) the time horizon is finite, and η, z have finite support, or (ii) regularity conditions on the contract, which are outlined in Lemma 3.

Our second proof in Section A.1.2 makes additional assumptions that permit reformulating the firm's problem using recursive contracts and a first-order approach (i.e., assuming that the incentive compatibility constraints may be summarized by the first-order condition to the worker's problem). The second proof is useful because it is closer to standard practice (e.g., Farhi and Werning, 2013) and because it derives results for the proof of Proposition 4 below.

Finally, after applying an envelope theorem, it is straightforward to derive the expression for the elasticity of market tightness in the flexible incentive pay economy with acyclical ex ante utility going to workers at the start of the contract, using similar steps to how we derived the impulse response of tightness in the rigid wage economy and equation (19).

A.1.1 Proof Environment 1: Sequence Problem

We seek to apply Theorem 4.13 of Bonnans and Shapiro (2000), which is reproduced below:

Bonnans and Shapiro (2000) Theorem 4.13 *Consider the following optimization problem:*

$$\min_{x \in \mathcal{X}} V(x, z) \quad \text{subject to } x \in \Phi$$

where z is a member of a Banach space Z , $\mathcal{X}$ is a Hausdorff topological space, $\Phi \subset \mathcal{X}$ is nonempty and closed, and $V : \mathcal{X} \times Z \rightarrow \mathbb{R}$ is continuous. Let the value function be defined as

$$J(z) \equiv \inf_{x \in \Phi(z)} V(x, z)$$

and the optimal control set be given by

$$\Gamma^*(z) \equiv \arg \min_{x \in \Phi(z)} V(x, z).$$

Suppose that $z_0 \in Z$ and

1. For all $x \in \mathcal{X}$ the function $V(x, \cdot)$ is Gateaux differentiable
2. $V(x, z)$ and its partial Fréchet derivative with respect to z , given by $D_z V(x, z)$, are continuous on $\mathcal{X} \times Z$
3. There exists $M \in \mathbb{R}$ and a compact set $C \subset \mathcal{X}$ such that for every z near z_0 the set $A(z) \equiv \{x \in \Phi : V(x, z) \leq M\}$ is non-empty and contained in C .

Then the optimal value function $J(\cdot)$ is Fréchet directionally differentiable at z_0 and

$$J'(z_0, d) = \inf_{x \in \Gamma^*(z_0)} D_z V(x, z_0) d,$$

where d is the direction of the Fréchet derivative and $J'(z_0, d)$ is the Fréchet derivative of J with respect to z in that direction.

This theorem provides conditions under which the total derivative of the value function with respect to some parameter z is equal to the partial derivative of the value function with respect to that parameter, taking the smallest product of the partial derivative and direction across the optimal control set. We verify the conditions of the theorem apply to the firm's problem, noting that the direction d corresponds to the sign of the increment $d \ln z_0$ in our one-dimensional context.

First, the space of possible aggregate productivities Z is clearly a Banach space, and the set of feasible contracts $\mathcal{X}$ is a Hausdorff topological space. By Assumption 1, Φ is non-empty. In addition, the firm's objective function $V(x, z)$ is continuous and is Gateaux differentiable since effort is assumed to continuously influence the measure of idiosyncratic profit shocks η . So, too, is its partial Fréchet derivative.

Thus, all that remains to be verified is: (i) the constraint set does not depend directly on z_0 and (ii) condition three of the theorem of Bonnans & Shapiro holds. To verify that the constraint set does not depend directly on z_0 , note that by inspection, neither the participation constraint (21) nor incentive constraints (22) depend directly on z_0 . Likewise, the bounds on w and a do not depend on z . Therefore, z does not directly enter the constraints, and the (B) term of equation (24) is zero.

Since Φ is compact, also by Assumption 1, we can verify condition 3 of Bonnans and Shapiro (2000) Theorem 4.13. In particular, setting $C = \Phi$ and $M = \max_{z \in [z, \bar{z}], x \in \Phi} V(x, z)$ verifies the condition. In this case, C is compact. We also have $A(z) = C = \Phi$ because all contracts x in Φ have a value of less than M .

We have now validated the conditions of Bonnans and Shapiro (2000) Theorem 4.13, and this envelope theorem applies to our problem.

We now apply the envelope theorem. Using the fact that z_0 is scalar we write the right-hand derivative as

$$\begin{aligned}
J'_+(z_0) &= \sup_{x^* \in \Gamma^*(z_0)} \frac{\partial}{\partial z_0} V(\mathbf{w}^*, \mathbf{a}^*; z_0) \\
&= \sup_{x^* \in \Gamma^*(z_0)} \frac{\partial}{\partial z_0} \left[\max_{\mathbf{w}(z_0), \mathbf{a}(z_0)} \sum_{t=0}^{\infty} (\beta(1-s))^t \int \int (f(\mathbb{E}[z_t|z_0] + \varepsilon_t, \eta_t) - w_t(\eta^t, \varepsilon^t; z_0)) \tilde{\pi}_t(\eta^t, \varepsilon^t | \mathbf{a}(z_0)) d\eta^t d\varepsilon^t \right] \\
&= \sup_{x^* \in \Gamma^*(z_0)} \left[\sum_{t=0}^{\infty} (\beta(1-s))^t \frac{\partial}{\partial z_0} \int \int (f(\mathbb{E}[z_t|z_0] + \varepsilon_t, \eta_t) - w_t(\eta^t, \varepsilon^t; z_0)) \tilde{\pi}_t(\eta^t, \varepsilon^t | \mathbf{a}(z_0)) d\eta^t d\varepsilon^t \right] \\
&= \sup_{x^* \in \Gamma^*(z_0)} \left[\sum_{t=0}^{\infty} (\beta(1-s))^t \frac{\partial}{\partial z_0} \int \int f(\mathbb{E}[z_t|z_0] + \varepsilon_t, \eta_t) \tilde{\pi}_t(\eta^t, \varepsilon^t | \mathbf{a}(z_0)) d\eta^t d\varepsilon^t \right] \quad (25) \\
&= \sup_{x^* \in \Gamma^*(z_0)} \left[\sum_{t=0}^{\infty} (\beta(1-s))^t \frac{\partial}{\partial z_0} \mathbb{E}[f(z_t, \eta_t) | \mathbf{a}(z_0)] \right]
\end{aligned}$$

where the second line substitutes in equation (20). Since f is continuously differentiable and Φ is compact, the supremum is attained at an optimum $x_+^* \in \Gamma^*$. Similarly, we have the left-hand derivative

$$J'_-(z_0) = \inf_{x^* \in \Gamma^*(z_0)} \left[\sum_{t=0}^{\infty} (\beta(1-s))^t \frac{\partial}{\partial z_0} \mathbb{E}[f(z_t, \eta_t) | z_0, \mathbf{a}^*] \right],$$

and the infimum is attained at an optimum $x_-^* \in \Gamma^*$. Combining the left- and right-hand derivatives, it follows that to a first-order

$$\begin{aligned}
dJ(z_0) &= \sup_{x^* \in \Gamma^*(z_0)} \left[\sum_{t=0}^{\infty} (\beta(1-s))^t \frac{\partial}{\partial z_0} \mathbb{E}[f(z_t, \eta_t) | z_0, \mathbf{a}^*] dz_0 \right] \\
&= \max_{x^* \in \Gamma^*(z_0)} \left[\sum_{t=0}^{\infty} (\beta(1-s))^t \frac{\partial}{\partial z_0} \mathbb{E}[f(z_t, \eta_t) | z_0, \mathbf{a}^*] dz_0 \right]
\end{aligned} \tag{26}$$

where if the increment dz_0 is negative, then, in effect, the supremum converts to an infimum, and the second line replaces the sup with a max because the space of optimal contracts is compact. Noting that the value of $J(z_0)$ is the same for all optimal contracts, the preceding equation implies

$$\begin{aligned}
\frac{dJ(z_0)}{J(z_0)} &= \frac{1}{\sum_{t=0}^{\infty} (\beta(1-s))^t \mathbb{E}[f(z_t, \eta_t) - w_t^* | z_0, \mathbf{a}^*]} \max_{x^* \in \Gamma^*(z_0)} \left[\sum_{t=0}^{\infty} (\beta(1-s))^t \frac{\partial}{\partial z_0} \mathbb{E}[f(z_t, \eta_t) | z_0, \mathbf{a}^*] dz_0 \right] \\
\Rightarrow d \ln J(z_0) &= \frac{\max_{x^* \in \Gamma^*(z_0)} \left[\sum_{t=0}^{\infty} (\beta(1-s))^t \frac{\partial}{\partial z_0} \mathbb{E}[f(z_t, \eta_t) | z_0, \mathbf{a}^*] z_0 \frac{dz_0}{z_0} \right]}{\sum_{t=0}^{\infty} (\beta(1-s))^t \mathbb{E}[f(z_t, \eta_t) - w_t^* | z_0, \mathbf{a}^*]} \\
&= \frac{\max_{x^* \in \Gamma^*(z_0)} \left[\sum_{t=0}^{\infty} (\beta(1-s))^t \frac{\partial}{\partial \ln z_0} \mathbb{E}[f(z_t, \eta_t) | z_0, \mathbf{a}^*] d \ln z_0 \right]}{\sum_{t=0}^{\infty} (\beta(1-s))^t \mathbb{E}[f(z_t, \eta_t) - w_t^* | z_0, \mathbf{a}^*]}.
\end{aligned}$$

The above equation and equation (17) then imply

$$\begin{aligned}
d \ln \theta_0 &= \frac{1}{\nu_0} d \ln J(z_0) \\
&= \frac{1}{\nu_0} \frac{\max_{x^* \in \Gamma^*(z_0)} \left[\sum_{t=0}^{\infty} (\beta(1-s))^t \frac{\partial}{\partial \ln z_0} \mathbb{E}[f(z_t, \eta_t) | z_0, \mathbf{a}^*] d \ln z_0 \right]}{\sum_{t=0}^{\infty} (\beta(1-s))^t \mathbb{E}[f(z_t, \eta_t) - w_t^* | z_0, \mathbf{a}^*]} \\
&= \frac{1}{\nu_0} \frac{\sum_{t=0}^{\infty} (\beta(1-s))^t \frac{\partial}{\partial \ln z_0} \mathbb{E}[f(z_t, \eta_t) | z_0, \mathbf{a}^*] d \ln z_0}{\sum_{t=0}^{\infty} (\beta(1-s))^t \mathbb{E}[f(z_t, \eta_t) - w_t^* | z_0, \mathbf{a}^*]}
\end{aligned}$$

where the last equality holds for some $(\mathbf{w}^*, \mathbf{a}^*) \in \Gamma^*(z_0)$. In particular, $(\mathbf{w}^*, \mathbf{a}^*)$ either maximizes the direct productivity effect among optimal contracts if $d \ln z_0$ is positive; or minimizes the direct productivity effect if $d \ln z_0$ is negative. We have derived equation (13) from the main text, characterizing the impulse response of tightness in the flexible incentive pay economy.

To prove the final part of the theorem, we now assume that the left- and right-hand partial derivatives of $dJ(z_0)$ are equal. A sufficient condition for this to hold is that the set

of optimal contracts is a singleton.¹⁵ We now derive the simplified expression for tightness dynamics in the neighborhood of the steady state, equation (15) from the main text. Starting from equation (25), we have

$$\begin{aligned}
J'(z_0) &= \max_{x^* \in \Gamma^*(z_0)} \left[\sum_{t=0}^{\infty} (\beta(1-s))^t \frac{\partial}{\partial z_0} \int \int f(\mathbb{E}[z_t|z_0] + \varepsilon_t, \eta_t) \tilde{\pi}_t(\eta^t, \varepsilon^t | \mathbf{a}^*(z_0)) d\eta^t d\varepsilon^t \right] \\
&= \max_{x^* \in \Gamma^*(z_0)} \left[\sum_{t=0}^{\infty} (\beta(1-s))^t \int \int \frac{\partial}{\partial z_0} f(\mathbb{E}[z_t|z_0] + \varepsilon_t, \eta_t) \tilde{\pi}_t(\eta^t, \varepsilon^t | \mathbf{a}^*(z_0)) d\eta^t d\varepsilon^t \right] \\
&= \max_{x^* \in \Gamma^*(z_0)} \left[\sum_{t=0}^{\infty} (\beta(1-s))^t \int \int f_z(z_t, \eta_t) \frac{\partial \mathbb{E}[z_t|z_0]}{\partial z_0} \tilde{\pi}_t(\eta^t, \varepsilon^t | \mathbf{a}^*(z_0)) d\eta^t d\varepsilon^t \right] \\
&= \max_{x^* \in \Gamma^*(z_0)} \left[\sum_{t=0}^{\infty} (\beta(1-s))^t \mathbb{E}[f_z(z_t, \eta_t) | \mathbf{a}^*(z_0)] \frac{\partial \mathbb{E}[z_t|z_0]}{\partial z_0} \right],
\end{aligned}$$

which, applying a similar reasoning to the derivation of equation (13) implies

$$\begin{aligned}
d \ln \theta_0 &= \frac{1}{\nu_0} \frac{\max_{x^* \in \Gamma^*(z_0)} \left[\sum_{t=0}^{\infty} (\beta(1-s))^t \mathbb{E}[f_z(z_t, \eta_t) | \mathbf{a}^*(z_0)] \frac{\partial \mathbb{E}[z_t|z_0]}{\partial \ln z_0} d \ln z_0 \right]}{\sum_{t=0}^{\infty} (\beta(1-s))^t \mathbb{E}[f(z_t, \eta_t) - w_t^* | z_0, \mathbf{a}^*]} \\
&= \frac{1}{\nu_0} \frac{\max_{x^* \in \Gamma^*(z_0)} \left[\sum_{t=0}^{\infty} (\beta(1-s))^t \mathbb{E}[f_z(z_t, \eta_t) | \mathbf{a}^*(z_0)] z_0 \frac{\partial \mathbb{E}[z_t|z_0]}{\partial z_0} d \ln z_0 \right]}{\sum_{t=0}^{\infty} (\beta(1-s))^t \mathbb{E}[f(z_t, \eta_t) - w_t^* | z_0, \mathbf{a}^*]} \\
&= \frac{1}{\nu_0} \frac{\max_{x^* \in \Gamma^*(z_0)} \left[\sum_{t=0}^{\infty} (\beta(1-s))^t \mathbb{E}[f(\eta_t) | \mathbf{a}^*(z_0)] z_0 d \ln z_0 \right]}{\sum_{t=0}^{\infty} (\beta(1-s))^t \mathbb{E}[f(z_t, \eta_t) - w_t^* | z_0, \mathbf{a}^*]} \\
&= \frac{1}{\nu_0} \frac{\max_{x^* \in \Gamma^*(z_0)} \left[\sum_{t=0}^{\infty} (\beta(1-s))^t \mathbb{E}[\bar{z} f(\eta_t) | \bar{z}, \mathbf{a}^*(\bar{z})] d \ln z_0 \right]}{\sum_{t=0}^{\infty} (\beta(1-s))^t \mathbb{E}[f(\bar{z}, \eta_t) - w_t^* | \bar{z}, \mathbf{a}^*(\bar{z})]} \\
&= \frac{1}{\nu_0} \frac{\max_{x^* \in \Gamma^*(z_0)} \left[\sum_{t=0}^{\infty} (\beta(1-s))^t \mathbb{E}[f(\bar{z}, \eta_t) | \bar{z}, \mathbf{a}^*(\bar{z})] d \ln z_0 \right]}{\sum_{t=0}^{\infty} (\beta(1-s))^t \mathbb{E}[f(\bar{z}, \eta_t) - w_t^* | \bar{z}, \mathbf{a}^*(\bar{z})]} \\
&= \frac{1}{\nu_0} \frac{\sum_{t=0}^{\infty} (\beta(1-s))^t \mathbb{E}[f(\bar{z}, \eta_t) | \bar{z}, \mathbf{a}^*(\bar{z})]}{\sum_{t=0}^{\infty} (\beta(1-s))^t \mathbb{E}[f(\bar{z}, \eta_t) - w_t^* | \bar{z}, \mathbf{a}^*(\bar{z})]} d \ln z_0 \\
&= \frac{1}{\nu_0} \frac{1}{1 - \frac{\sum_{t=0}^{\infty} (\beta(1-s))^t \mathbb{E}[w_t^* | \bar{z}, \mathbf{a}^*(\bar{z})]}{\sum_{t=0}^{\infty} (\beta(1-s))^t \mathbb{E}[f(\bar{z}, \eta_t) | \bar{z}, \mathbf{a}^*(\bar{z})]}} d \ln z_0 \\
&\implies \frac{d \ln \theta_0}{d \ln z_0} = \frac{1}{\nu_0} \frac{1}{1 - \frac{\sum_{t=0}^{\infty} (\beta(1-s))^t \mathbb{E}[w_t^* | \bar{z}, \mathbf{a}^*(\bar{z})]}{\sum_{t=0}^{\infty} (\beta(1-s))^t \mathbb{E}[f(\bar{z}, \eta_t) | \bar{z}, \mathbf{a}^*(\bar{z})]}}; \tag{27}
\end{aligned}$$

The third line uses the fact that $\frac{\partial \mathbb{E}[z_t|z_0]}{\partial z_0} = 1$ because z_t follows a driftless random walk and

¹⁵When the left- and right-hand derivatives of $dJ(z_0)$ are different, we can still derive tightness dynamics for negative and positive shocks in the neighborhood of the steady state.

$f_z(z_t, \eta_t) = \eta_t$ because $f(z_t, \eta_t)$ is homogeneous of degree one in z_t ; the fourth line evaluates derivatives at the non-stochastic state in which $z_t = \bar{z}$; and the sixth line uses the uniqueness of the optimal contract. Equation (27) is the same as equation (15) from the main text for the case of the flexible incentive pay economy. The derivation of equation (27) for the case of rigid wages is virtually identical, so we do not repeat it here. This derivation completes the proof of Theorem 1. $\square$

A.1.2 Proof Environment 2: First-Order Approach and Recursive Formulation

We now pursue the First-Order Approach (FoA). To make progress, we must introduce one additional assumption guaranteeing that the so-called FoA offers a valid solution to the contracting problem:

Assumption 2. *The set of feasible contracts $(\mathbf{w}, \mathbf{a}) \in \mathcal{X}$ is compact and convex. Assume standard Inada conditions on utility, $\lim_{c \rightarrow \underline{w}} u_c(c, a) = \lim_{a \rightarrow \bar{a}} u_c(c, a) = \infty$ and $\lim_{c \rightarrow \bar{w}} u_c(c, a) = \lim_{a \rightarrow \underline{a}} u_c(c, a) = 0$, that the worker's optimal effort choices are determined by the first-order condition to problem (6) and that the density of η_t can be expressed as*

$$\pi_t(\eta_t | \eta^{t-1}, a^t) = \pi_t(\eta_t | \eta_{t-1}, a_t).$$

Under this assumption, the incentive compatibility constraint may be written as the first-order condition to the worker's problem, and the firm's contracting problem may be expressed recursively.

We now show how to apply an envelope theorem to the flexible incentive pay problem under this stronger assumption. This proof is clarifying because the approach is closer to standard practice, and it will also be useful because it derives results that are necessary for Proposition 4. Therefore, for this subsection, we make both Assumptions 1 and 2 from the main text.

The application of the envelope theorem proceeds in three steps in this environment. First, we derive a first-order approach to simplify incentive constraints into local incentive constraints as in Farhi and Werning (2013) or Pavan et al. (2014). Then, we develop a recursive formulation of the problem following the steps in Ndiaye (2020). Finally, we use these constructions to prove our main theorem.

Step 1: First-Order Approach The first-order condition for a_t in the worker's problem (6) given a contract is

$$0 = \int \int \left[u_a \left(w_t(\eta^t, z^t), a_t(\eta^{t-1}, z^t) \right) \tilde{\pi}_t(\eta^t, z^t | z_0, \mathbf{a}) + u \left(w_t(\eta^t, z^t), a_t(\eta^{t-1}, z^t) \right) \frac{\partial}{\partial a_t} \tilde{\pi}_t(\eta^t, z^t | z_0, \mathbf{a}) \right] d\eta^t dz^t$$

Note that this holds for every t and realization of z^t . Thus, one can remove the outer integral to write first-order incentive constraints as

$$\int \left[u_a \left(w_t(\eta^t, z^t), a_t(\eta^{t-1}, z^t) \right) \pi_t(\eta_t | z^t, \eta^{t-1}, \mathbf{a}) + u \left(w_t(\eta^t, z^t), a_t(\eta^{t-1}, z^t) \right) \frac{\partial}{\partial a_t} \pi_t(\eta_t | z^t, \eta^{t-1}, \mathbf{a}) \right] d\eta_t = 0$$

Step 2: Recursive Formulation We will work with the relaxed problem and develop a recursive formulation of the firm's problem. Notationally, let the value of some variable X in the period t problem be given by X , the value of X in $t - 1$ be given as X_- , and the value of X in $t + 1$ be given by X' . Suppressing explicit dependence of the problem on initial productivity z_0 for notational convenience, the recursive formulation of the firm's problem is then (we now drop the history dependence with the assumption that the process for η is a Markov process):

$$J(v_-, \eta_-, z_-, t) = \max_{a(\eta_-, z), w(\eta, z), v(\eta, z)} \int \int \left[f(\eta, z) - w(\eta, z) + \beta(1-s) J(v(\eta, z), \eta, z, t+1) \right] \pi(\eta | z, \eta_-, a(\eta_-, z)) \hat{\pi}(z | z_-) d\eta dz \quad (28)$$

subject to the following constraints:

$$\omega(\eta, z) = u(w(\eta, z), a(\eta_-, z)) + \beta \left[(1-s) v(\eta, z) + s \int U(z') \hat{\pi}(z' | z) dz' \right] \quad (29)$$

for all η and z realizations,

$$[\lambda] : \quad v_- \leq \int \int \omega(\eta, z) \pi(\eta | z, \eta_-, a(\eta_-, z)) \hat{\pi}(z | z_-) d\eta dz, \quad (30)$$

and the first-order incentive constraints:

$$\int \left[u_a \left(w(\eta, z), a(\eta_-, z) \right) \pi(\eta | z, \eta_-, a) + u \left(w(\eta, z), a(\eta_-, z) \right) \frac{\partial}{\partial a} \pi(\eta | z, \eta_-, a) \right] d\eta = 0. \quad (31)$$

The firm begins period t knowing the prior realization of shocks z_- and η_- and inherits a utility it must promise to the worker over the remaining life of the contract, which we denote v_- . The firm's flow profits are the expected output $f(\eta, z)$ minus their expected wage payments $w(\eta, z)$. Firms additionally receive a continuation value with probability $1 - s$, which they discount at rate β . The firm maximizes the sum of flow profits and continuation values by choosing the suggested effort and wage functions for every realization of η and z , as well as a function for the next period's promised utility to the worker $v(\eta, z)$, subject to some constraints that we now describe.

The worker's value under the contract given a realization (η, z) is given by $\omega(\eta, z)$, defined in equation (29). It is equal to the worker's flow utility $u(w(\eta, z), a(\eta_-, z))$ plus a continuation value. With probability s , the match dissolves, and the worker receives the value of unemployment. With probability $1 - s$, the match survives, and the worker receives $v(\eta, z)$.

The recursive version of the participation constraint states that the worker's expected value under the contract must be at least the value promised to them v , and is given by equation (30). Note that v_- in the initial period of the match maps to the utility promised to the worker overall $\mathcal{B}$ in the non-recursive formulation of the problem. For periods after the start of the contract, equation (30) may be interpreted as a promise-keeping constraint. Equation (31) is the relaxed incentive constraint described above.

Let the Lagrangian of the recursive problem be defined by $\int \int \mathcal{L}(\cdot) d\eta dz$ for

$$\begin{aligned} \mathcal{L} \equiv & [f(\eta, z) - w(\eta, z; z_0)] \pi(\eta|z, \eta_-, a(\eta_-, z)) \hat{\pi}(z|z_-) \\ & + \beta(1 - s) \left[J(v(\eta, z), \eta, z, t + 1) \right] \pi(\eta|z, \eta_-, a(\eta_-, z)) \hat{\pi}(z|z_-) \\ & - \lambda [v_- - \omega(\eta, z) \pi(\eta|z, \eta_-, a(\eta_-, z)) \hat{\pi}(z|z_-)] \\ & - \gamma(z) \left[u_a \left(w(\eta, z), a(\eta_-, z) \right) \pi(\eta|z, \eta_-, a) + u \left(w(\eta, z), a(\eta_-, z) \right) \frac{\partial}{\partial a} \pi(\eta|z, \eta_-, a) \right], \end{aligned} \quad (32)$$

where λ is the Lagrange multiplier on the participation constraint and $\gamma(z)$ is the multiplier on the incentive constraint given aggregate productivity z . Again, we suppress dependence on z_0 , but the firm's choice variables and the distribution of z and η may all depend on z_0 .

Next, we introduce the change of variable with the notation $z_t = \mathbb{E}[z_t|z_0] + \varepsilon_t$, where by definition, ε_t is the cumulative innovation to the process for z between 0 and t and ε_0 is known to be 0. One can write the Lagrangian as:

$$\begin{aligned} \mathcal{L} = & [f(\eta, \mathbb{E}[z|z_0] + \varepsilon) - w(\eta, \varepsilon)] \pi(\eta|\varepsilon, \eta_-, a(\eta_-, \varepsilon)) \hat{\pi}(\varepsilon|\varepsilon_-) \\ & + \beta(1 - s) \left[J(v(\eta, \varepsilon), \eta, \varepsilon, t + 1) \right] \pi(\eta|\varepsilon, \eta_-, a(\eta_-, \varepsilon)) \hat{\pi}(\varepsilon|\varepsilon_-) \\ & - \lambda [v_- - \omega(\eta, \varepsilon) \pi(\eta|\varepsilon, \eta_-, a(\eta_-, \varepsilon)) \hat{\pi}(\varepsilon|\varepsilon_-)] \\ & - \gamma(\varepsilon) \left[u_a \left(w(\eta, \varepsilon), a(\eta_-, \varepsilon) \right) \pi(\eta|\varepsilon, \eta_-, a) + u \left(w(\eta, \varepsilon), a(\eta_-, \varepsilon) \right) \frac{\partial}{\partial a} \pi(\eta|\varepsilon, \eta_-, a) \right] \end{aligned} \quad (33)$$

Step 3: Envelope Theorem We seek to apply Theorem 1 of Marimon and Werner (2021), which relies on the following technical assumptions.

Technical Assumptions:

TA1. The set $\mathcal{X}$ of feasible allocations is convex, and f, u, π, u_a , and π_a are continuous functions of $\{a, w, z_0\}$

TA2. The constraint set $\mathcal{G}(z_0) = \{(\mathbf{w}, \mathbf{a}) \in \mathcal{X} : G(\mathbf{w}, \mathbf{a}; z_0) \leq 0\}$ is compact for every $z \in Z$, a neighborhood of z_0 , and there exists a contract $(\mathbf{w}, \mathbf{a})$ such that the participation constraint (30) is slack.

TA3. The set of optimal contracts is non-empty.

We argue these conditions apply in our setting. $\mathcal{X}$ is convex as the product of segments. Under Assumption 1, $\mathcal{X}$ is compact. Then the constraint set $\mathcal{G}(z_0)$ is a closed subset of a compact and so is compact. What's more, there exists a contract such that the participation constraint is slack since, for every z_0 and promised utility v_- , there exists a feasible continuation value and effort $\omega(\eta, z), a(\eta, z)$ that yield strictly higher utility than v_- : that is inequality (30) is strict. Finally, since from Assumption 1, $\mathcal{X}$ is compact and non-empty, the set of optimizers of our continuous objective (i.e., the set of optimal contracts) is non-empty.¹⁶

One can now apply the envelope theorem of Marimon and Werner (2021) to argue that the derivative of the value function with respect to all variables the firm chooses and costates – a^*, w^*, v^*, λ^* , and γ^* – sum to zero. Therefore, differentiating the Lagrangian (33) with respect to z_0 and substituting in for $\omega(\eta, \varepsilon)$ yields the right-hand derivative:

$$\begin{aligned} \frac{\partial J(v_-, \eta_-, z_-, t)}{\partial z_0^+} &= \sup_{(\mathbf{w}^*, \mathbf{a}^*) \in \Gamma^*(z_0)} \int \int \frac{\partial}{\partial z_0} [f(\eta, \mathbb{E}[z|z_0] + \varepsilon)] \pi(\eta|\varepsilon, \eta_-, a^*(\eta_-, \varepsilon)) \hat{\pi}(\varepsilon|\varepsilon_-) d\eta d\varepsilon \\ &+ \beta(1-s) \int \int \frac{\partial}{\partial z_0^+} \left[J(v^*(\eta, \varepsilon), \eta, \varepsilon, t+1) \right] \pi(\eta|\varepsilon, \eta_-, a^*) \hat{\pi}(\varepsilon|\varepsilon_-) d\eta d\varepsilon \\ &+ \beta s \lambda^*(\eta_-, z_-) \int \int \frac{\partial}{\partial z_0} U(\mathbb{E}[z'|z_0] + \varepsilon') \hat{\pi}(\varepsilon'|\varepsilon) \hat{\pi}(\varepsilon|\varepsilon_-) d\varepsilon' d\varepsilon. \end{aligned} \tag{34}$$

This is a refinement of a recursive version of equation (24): the first-order impact of aggregate productivity on the value of a filled job is given by the sum of the direct effect on the firm's flow and continuation values, plus the direct effect on the constraints. Two terms are missing from the fuller decomposition in equation (24). First, the “B-term” features no direct effect on incentive constraints. This arises from the assumption that the distribution of η and ε

¹⁶This envelope theorem is better suited for our purposes than Corollary 5 of Milgrom and Segal (2002) since it does not require compactness assumptions on the support of the shocks.

do not directly depend on z_0 . Second, the “C-term” – the indirect effect on firm value that arises from changes in the contracted wages or effort – does not appear because we have applied the envelope theorem of Marimon and Werner (2021).

We can write explicitly the sequence of participation constraints from time 0 as:

$$\begin{aligned} \lambda_-(z_0) : \quad & \mathcal{E}(z_0) \leq v \\ \lambda_{t-1}(\eta^{t-1}, z^{t-1}) : \quad & v_{t-1}(\eta^{t-1}, z^{t-1}) \leq \int \int \omega(\eta^t, z^t) \pi(\eta_t | z^t, \eta^{t-1}, a(\eta^{t-1}, z^t)) \hat{\pi}(z_t | z^{t-1}) d\eta_t dz_t, \forall t \geq 1. \end{aligned}$$

The corresponding sequential participation constraints are:

$$\begin{aligned} [\lambda_-(z_0)] : \quad & \mathcal{B}(z_0) \leq \sum_{t=0}^{\infty} (\beta(1-s))^t \left[\int \int u(w_t(\eta^t, \varepsilon^t; z_0), a_t(\eta^{t-1}, \varepsilon^t; z_0)) \tilde{\pi}_t(\eta^t, \varepsilon^t | \mathbf{a}(z_0)) d\eta^t d\varepsilon^t \right. \\ & \left. + \beta s \int U(\mathbb{E}[z_{t+1} | z_0] + \varepsilon_{t+1}) \hat{\pi}_{t+1}(\varepsilon^{t+1}) d\varepsilon^{t+1} \right] \\ [\lambda_\tau(\eta^\tau, \varepsilon^\tau; z_0)] : \quad & \sum_{t=\tau+1}^{\infty} (\beta(1-s))^{t-\tau-1} \left[\int \int u(w_t(\eta^t, \varepsilon^t; z_0), a_t(\eta^{t-1}, \varepsilon^t; z_0)) \tilde{\pi}_t(\eta^t, \varepsilon^t | \mathbf{a}(z_0)) d\eta^t d\varepsilon^t \right. \\ & \left. + \beta s \int U(\mathbb{E}[z_{t+1} | z_0] + \varepsilon_{t+1}) \hat{\pi}_{t+1}(\varepsilon^{t+1}) d\varepsilon^{t+1} \right] \geq v_\tau(\eta^\tau, \varepsilon^\tau; z_0), \quad \forall \tau = 0, \dots, +\infty \end{aligned} \quad (35)$$

Now we apply the envelope theorem to the problem recursively, replacing $\mathcal{E}$ with its equilibrium value $\mathcal{B}$ to obtain

$$\begin{aligned} \frac{\partial J}{\partial z_0} = & \sup_{\{w^*, a^*\} \in \Gamma^*(z_0)} \sum_{t=0}^{+\infty} \left[\int \int (\beta(1-s))^t \frac{\partial f(\mathbb{E}[z_t | z_0] + \varepsilon_t, \eta_t)}{\partial z_0} \tilde{\pi}_t(\eta^t, \varepsilon^t | \mathbf{a}(z_0)) d\eta^t d\varepsilon^t \right. \\ & - \lambda_-(z_0) \left[\frac{\partial \mathcal{B}(z_0)}{\partial z_0} - \beta s \sum_{t=1}^{+\infty} \int \int (\beta(1-s))^{t-1} \frac{\partial U(\mathbb{E}[z_t | z_0] + \varepsilon_t)}{\partial z_0} \hat{\pi}(\varepsilon^t) d\varepsilon^t \right] \\ & + \sum_{\tau=0}^{\infty} \int \int \lambda_\tau(\eta^\tau, \varepsilon^\tau; z_0) \beta s \left[\sum_{t=\tau+2}^{+\infty} \int \int (\beta(1-s))^{t-\tau-2} \frac{\partial U(\mathbb{E}[z_t | z_0] + \varepsilon_t)}{\partial z_0} \hat{\pi}(\varepsilon^t | \varepsilon^\tau) d\varepsilon^t \right] \times \\ & \left. \tilde{\pi}_\tau(\eta^\tau, \varepsilon^\tau | \mathbf{a}(z_0)) d\eta^\tau d\varepsilon^\tau \right] \end{aligned} \quad (36)$$

When the outside option of the worker is acyclical and TIOLI, we have:

$$\frac{\partial J(z_0)}{\partial z_0} = \sup_{\{w^*, a^*\} \in \Gamma^*(z_0)} \sum_{t=0}^{\infty} (\beta(1-s))^t \left[\int \int \frac{\partial f(\mathbb{E}[z_t | z_0] + \varepsilon_t, \eta_t)}{\partial z_0} \tilde{\pi}_t(\eta^t, \varepsilon^t | \mathbf{a}^*(z_0)) d\eta^t d\varepsilon^t \right]. \quad (37)$$

The preceding equation is the same as equation (26) from A.1.1, the previous application of

the envelope theorem. Therefore, the same manipulations performed at the end of Section A.1.1 yield the market tightness dynamics in Theorem 1. $\square$

A.2 Sufficient Conditions for Compactness of Φ

This section provides two sets of sufficient conditions for Φ to be compact. The first condition is that η and z have finite support, and contracts last at least T periods for T finite. The second is that contracts are continuous and twice differentiable in their arguments with uniformly bounded first and second derivatives and that T is finite.

Lemma 2. *If η and z have finite support and the time horizon is finite, then the choice set of contracts is compact, and the envelope theorem holds.*

Proof. Suppose $\eta_t \in \{\eta_1, \dots, \eta_N\}$ and $z_t \in \{z_1, \dots, z_M\}$ have finite support and T is finite. We will show that the space of $[w, a]$ functions of (η, z) that are IC and PC is compact.

Consider a sequence of functions $[w_n, a_n]$ that are IC and satisfy the PC. The sequence $[w_n(\eta_1, z_1), a_n(\eta_1, z_1)]$ takes values in a compact set. Therefore, it has a subsequence that converges. Call it $[w_{\phi_{1,1}(n)}(\eta_1, z_1), a_{\phi_{1,1}(n)}(\eta_1, z_1)]$. Now apply the same reasoning to the sequence $[w_{\phi_{1,1}(n)}(\eta_1, z_2), a_{\phi_{1,1}(n)}(\eta_1, z_2)]$; similarly, it is in a compact set, so it has a subsequence that converges.

Through this diagonal argument, we construct $[w_{\phi_{1,1} \circ \phi_{1,2} \dots \phi_{N,M}(n)}, a_{\phi_{1,1} \circ \phi_{1,2} \dots \phi_{N,M}(n)}]$: a sub-sequence of *functions* that converges. Now we need to show that the limiting function is also in the set, that is, it is IC and satisfies PC.

The PC is a closed inequality involving continuous functions. Due to the continuity of the involved functions, the limit of any sequence of functions satisfying the PC will also satisfy it. Analogously, for the incentive compatibility constraint (IC), consider fixing an action $\tilde{a}(\cdot)$. Any sequence of $[w, a]$ satisfying the IC inequality for $\tilde{a}(\cdot)$, by the continuity of the functions involved, will satisfy it at the limit. Since this applies for all $\tilde{a}(\cdot)$, the limiting function must be IC. We began with an arbitrary sequence of $[w, a]$ that are both IC and PC, and we have shown that it has a subsequence converging to a limit that is also IC and PC. Therefore, the space of mechanisms is compact.

We can now employ a standard envelope theorem in this case, given that the choice set is compact and Corollary 4 of Milgrom and Segal (2002) applies. $\square$

Lemma 3. *The set of feasible contracts that satisfy the IC constraints, Φ , is compact if contracts are restricted to being continuous and twice differentiable in their arguments $\{\eta^t, z^t\}$, with uniformly bounded first and second derivatives and the horizon T is finite.*

Proof. We will show that Φ is equicontinuous.¹⁷ Let $\Xi = [\underline{\eta}, \bar{\eta}]$ be the set of possible values for η . Consider a set of functions that are continuously differentiable on $[0, 1]$ and such that both the functions and their first and second derivatives are uniformly bounded. This means there exists some real number M such that for every function f in the set and every $x \in \Xi^T \times Z^T$, $\|f(x)\| \leq M$ and for its Jacobian $\|Df(x)\| \leq M$, where $\|\cdot\|$ is the Euclidian norm in $\Xi^T \times Z^T$.

Given $\epsilon > 0$, choose $\delta = \epsilon/2M$. Then for any function f in Φ and any points x and y in $\Xi^T \times Z^T$ such that $\|x - y\| < \delta$, by the mean value theorem, we have $\|f(x) - f(y)\|_\infty = |Df(c)| \cdot \|x - y\|$ for some c in the line $xt + (1 - t)y$, $t \in [0, 1]$. Since $|Df(c)| \leq M$ and $\|x - y\| < \delta = \epsilon/2M$, we get $\|f(x) - f(y)\|_\infty < \epsilon/2$.

Similarly, we can apply the mean value theorem to the Jacobian of f , and since the second derivatives are bounded, an analogous argument to that above yields $\|Df(x) - Df(y)\|_\infty < \epsilon/2$. Therefore $\|f(x) - f(y)\|_{C^1} \equiv \|f(x) - f(y)\|_\infty + \|Df(x) - Df(y)\|_\infty < \epsilon$ and we have shown that Φ is equicontinuous. By the Ascoli Theorem, any sequence in Φ thus has a subsequence that converges. Therefore, Φ is compact. $\square$

A.3 Non-incentive reasons for wage cyclicalty.

To derive Theorem 1, we assumed that workers' promised utility is acyclical. This assumption provides the clean benchmark in which all wage cyclicalty in the model arises due to incentives. We now relax this assumption to consider non-incentive reasons for wage cyclicalty. To do so, we assume that the ex ante utility promised to the worker $\mathcal{E}(z_0)$ is given by a reduced-form function $\mathcal{B}(z_0)$: firms commit to providing workers with utility $\mathcal{B}(z_0)$ over the contract.¹⁸ Common bargaining protocols in the labor search literature implicitly define different functions for $\mathcal{B}(z_0)$. For instance, if firms make take-it-or-leave-it offers to workers, the value of employment is equal to the value of nonemployment: $\mathcal{B}(z_0) = \sum_t \beta^t \mathbb{E}[\xi(z_t)|z_0]$, where $\xi(z_t)$ is the flow value of unemployment. This nests the case in which unemployment benefits or the opportunity cost of unemployment is procyclical (Hagedorn et al., 2013; Chodorow-Reich and Karabarbounis, 2016; Mitman and Rabinovich, 2019). Nash bargaining also implicitly defines an increasing function for $\mathcal{B}(z_0)$, as we prove below. Our formulation also evokes a notion of unemployment as a “worker discipline device” (Shapiro and Stiglitz, 1984): If the value of employment is low because unemployment at present or in the future is costly, workers will offer higher effort at lower wages.

¹⁷ Φ is said to be equicontinuous at a point $x \in \Xi^T \times Z^T$ if, for every $\epsilon > 0$, there exists a $\delta > 0$ such that for every function f in Φ and every point y in $\Xi^T \times Z^T$, if $\|x - y\| < \delta$ then $\|f(x) - f(y)\|_{C^1} < \epsilon$, where $\|\cdot\|_{C^1}$ is the C^1 norm.

¹⁸See Blanchard and Galí (2010) and Michaillat (2012) for an implementation of this approach in search models without effort.

The reduced-form approach has two advantages. First, our conclusions about the role of bargaining and outside options will be robust to a specific protocol. Second, we can tractably incorporate bargaining into dynamic incentive contract models. The disadvantage of the approach is that $\mathcal{B}(z_0)$ is a reduced-form object, which is not invariant to changes in the primitives of the environment.

Let $\mathcal{W}(z_0)$ denote the present discounted value of wage payments under the optimal wage contract:

$$\mathcal{W}(z_0) \equiv \sum_{t=0}^{\infty} (\beta(1-s))^t \int w_t^*(\eta^t, z^t) \tilde{\pi}_t(\eta^t, z^t | z_0, \mathbf{a}^*(z_0)) d\eta^t dz^t.$$

Likewise, Let $\mathcal{Y}(\mathbf{a}^*(z_0), z_0)$ denote the expected present discounted value of output from a match given the optimal effort function $\mathbf{a}^*(z_0)$:

$$\mathcal{Y}(\mathbf{a}^*(z_0), z_0) \equiv \sum_{t=0}^{\infty} (\beta(1-s))^t \int \int f(z_t, \eta_t) \tilde{\pi}_t(\eta^t, z^t | z_0, \mathbf{a}^*(z_0)) d\eta^t dz^t.$$

Denote as $\tilde{\mathcal{B}}(z)$ the ex ante utility promised to the worker at the start of the contract, net of her continuation value with regard to separation into unemployment: $\tilde{\mathcal{B}}(z) \equiv \mathcal{B}(z) - \sum_{t=0}^{\infty} (\beta(1-s))^t \beta s \mathbb{E}[U(z_{t+1}) | z_0 = z]$. Fluctuations in $\tilde{\mathcal{B}}(z)$ capture variations in workers' ex ante utility due to, for instance, bargaining power or changes in worker outside options. Finally, define *non-incentive wage cyclical* (*NWC*) as wage movements over and above output movements driven by effort fluctuations

$$\frac{\partial \mathcal{W}^{\text{non-incentive}}(z_0)}{\partial z_0} \equiv \frac{d\mathcal{W}(z_0)}{dz_0} - \partial_{\mathbf{a}} \mathcal{Y}(\mathbf{a}^*(z_0); z_0) \frac{d\mathbf{a}^*}{dz_0}. \quad (38)$$

Note that Theorem 1 states that non-incentive wage cyclical is zero under Assumption 1 when $\tilde{\mathcal{B}}(z_0)$ is acyclical; this is the sense in which $\frac{\partial \mathcal{W}^{\text{non-incentive}}(z_0)}{\partial z_0}$ is not related to incentives.

Characterizing the response of market tightness to productivity becomes more difficult when $\tilde{\mathcal{B}}(z)$ is nonconstant, as the set of contracts satisfying the participation constraint now moves directly with z_0 . We therefore must maintain Assumption 2 in Appendix Section A.1.2 so that the FoA offers a valid solution to the contracting problem. This then yields the following proposition:

Proposition 4. *Assume that Assumptions 1 and 2 hold. The impulse response of market tightness to aggregate shocks in the flexible incentive pay economy is*

$$d \ln \theta_0 = \frac{1}{\nu_0} \frac{\sum_{t=0}^{\infty} (\beta(1-s))^t \frac{\partial}{\partial \ln z_0} \mathbb{E}[f(z_t, \eta_t) | z_0, \mathbf{a}^*(z_0)] - \frac{\partial \mathcal{W}^{\text{non-incentive}}(z_0)}{\partial \ln z_0}}{\sum_{t=0}^{\infty} (\beta(1-s))^t \mathbb{E}[f(z_t, \eta_t) - w_t^*(z_0) | z_0, \mathbf{a}^*(z_0)]} d \ln z_0, \quad (39)$$

where $\partial \mathcal{W}^{\text{non-incentive}}(z_0)/\partial \ln z_0$ is defined in equation (38). Moreover,

$$\frac{\partial \mathcal{W}^{\text{non-incentive}}(z_0)}{\partial \ln z_0} > 0 \quad \Longleftrightarrow \quad \tilde{\mathcal{B}}'(z_0) > 0;$$

i.e., *Non-Incentive Wage Cyclicalities* is positive if and only if *ex ante utility* is procyclical.

Equation (39) shows that Non-Incentive Wage Cyclicalities (NWC) appears alongside the direct productivity effect. Therefore, when NWC is high, the impulse response of tightness is small – wage cyclicalities arising for non-incentive reasons dampen employment responses to shocks as in the standard model. The proposition also shows that what we have defined as NWC corresponds to the cyclicalities of workers' ex ante utility—NWC is positive if and only if the utility promised to workers at the start of a contract is procyclical.

A.3.1 Proof of Proposition 4

First, we derive equation (39) from the main text. Differentiating the value of a filled job $J(z_0)$ with respect to z_0 yields the following expression:

$$\frac{dJ(z_0)}{dz_0} = \frac{\partial \mathcal{Y}(\mathbf{a}^*(z_0); z_0)}{\partial z_0} - \left(\frac{d\mathcal{W}(z_0)}{dz_0} - \partial_{\mathbf{a}} \mathcal{Y}(\mathbf{a}^*(z_0); z_0) \frac{d\mathbf{a}^*}{dz_0} \right). \quad (40)$$

and from equation (38) we have

$$\frac{\partial \mathcal{W}^{\text{non-incentive}}(z_0)}{\partial z_0} \equiv \frac{d\mathcal{W}(z_0)}{dz_0} - \partial_{\mathbf{a}} \mathcal{Y}(\mathbf{a}^*(z_0); z_0) \frac{d\mathbf{a}^*}{dz_0}.$$

The preceding two equations imply

$$\begin{aligned} \frac{dJ(z_0)}{dz_0} &= \frac{\partial \mathcal{Y}(\mathbf{a}^*(z_0); z_0)}{\partial z_0} - \frac{\partial \mathcal{W}^{\text{non-incentive}}(z_0)}{\partial z_0} \\ &= \sum_{t=0}^{\infty} (\beta(1-s))^t \frac{\partial}{\partial z_0} \mathbb{E}[f(z_t, \eta_t) | z_0, \mathbf{a}^*] - \frac{\partial \mathcal{W}^{\text{non-incentive}}(z_0)}{\partial z_0} \\ \implies \frac{d \ln J(z_0)}{d \ln z_0} &= \frac{z_0 \left(\sum_{t=0}^{\infty} (\beta(1-s))^t \frac{\partial}{\partial z_0} \mathbb{E}[f(z_t, \eta_t) | z_0, \mathbf{a}^*] - \frac{\partial \mathcal{W}^{\text{non-incentive}}(z_0)}{\partial z_0} \right)}{\sum_{t=0}^{\infty} (\beta(1-s))^t \mathbb{E}[f(z_t, \eta_t) - w_t^* | z_0, \mathbf{a}^*]} \\ &= \frac{\sum_{t=0}^{\infty} (\beta(1-s))^t \frac{\partial}{\partial \ln z_0} \mathbb{E}[f(z_t, \eta_t) | z_0, \mathbf{a}^*] - \frac{\partial \mathcal{W}^{\text{non-incentive}}(z_0)}{\partial \ln z_0}}{\sum_{t=0}^{\infty} (\beta(1-s))^t \mathbb{E}[f(z_t, \eta_t) - w_t^* | z_0, \mathbf{a}^*]}, \end{aligned}$$

which is equation (39) from the main text.

Now, we are going to prove the “moreover” statement that NWC is positive if and only if the promised utility is procyclical. The derivation makes use of equation (36) derived in Section A.1.2. Suppose the optimal contract features optimal choices for wages and effort, which are in the interior of Φ , which is true under the Inada conditions made in Assumption 2. Then the additional Lagrangian terms after time zero are non-binding, and equation (36) becomes

$$\begin{aligned} \frac{\partial J}{\partial z_0^+} = & \sup_{\{w^*, a^*\} \in \Gamma^*(z_0)} \sum_{t=0}^{\infty} \left[\int \int (\beta(1-s))^t \frac{\partial f(\mathbb{E}[z_t|z_0] + \varepsilon_t, \eta_t)}{\partial z_0} \tilde{\pi}_t(\eta^t, \varepsilon^t | \mathbf{a}^*(z_0)) d\eta^t d\varepsilon^t \right. \\ & \left. - \lambda_{PC}^*(z_0) \left[\frac{\partial \mathcal{B}(z_0)}{\partial z_0} - \beta s \sum_{t=1}^{+\infty} \int \int (\beta(1-s))^{t-1} \frac{\partial U(\mathbb{E}[z_t|z_0] + \varepsilon_t)}{\partial z_0} \hat{\pi}(\varepsilon^t) d\varepsilon^t \right] \right] \end{aligned} \quad (41)$$

Under Assumption 1, $\mathcal{X}$ is compact, and so the supremum is achieved at a contract $\{w^*, a^*\} \in \Gamma^*(z_0)$. Evaluated at that optimum and comparing equation (39) with equation (41) yields

$$\lambda_{PC}^*(z_0) \left[\frac{\partial \tilde{\mathcal{B}}(z_0)}{\partial z_0} \right] = \frac{\partial \mathcal{W}^{non-incentive}(z_0)}{\partial z_0}.$$

Finally, $\lambda_{PC}^*(z_0) > 0$ because the participation constraint must bind on the optimal contract. It immediately follows that $\partial \mathcal{W}^{non-incentive}(z_0)/\partial z_0 > 0$ if and only if $\partial \tilde{\mathcal{B}}(z_0)/\partial z_0 > 0$. $\square$

A.3.2 Implicit Definition of $\mathcal{B}(z_0)$ under Nash Bargaining

This subsection shows that Nash bargaining implicitly defines a functional form for $\mathcal{B}(z_0)$. Suppose that the firm and worker engage in generalized Nash bargaining over the surplus of the match, and φ is the firm’s bargaining power. Firms and workers take as given the utility that workers would receive were they to match with another firm next period $\mathcal{E}(z)$. Promised utility $\mathcal{B}(z_0)$ is implicitly defined by

$$\mathcal{B}(z_0) = \arg \max_{\bar{\mathcal{B}}} J(z_0, \bar{\mathcal{B}})^\varphi (\bar{\mathcal{B}} - U(z_0))^{1-\varphi}.$$

Here, as in the main text, $U(z_0)$ is the value of unemployment at time 0. $J(z_0, \bar{\mathcal{B}})$ is defined by equations (5)-(7) in the main text, replacing $\mathcal{E}(z_0)$ with $\bar{\mathcal{B}}$ in equation (7). Therefore, $\mathcal{B}(z_0)$ is the solution to the standard Nash bargaining problem, albeit in an environment with dynamic incentive pay. The solution is

$$\varphi \frac{\frac{\partial J(z_0, \mathcal{B}(z_0))}{\partial \bar{\mathcal{B}}}}{J(z_0, \mathcal{B}(z_0))} + \frac{(1-\varphi)}{\mathcal{B}(z_0) - U(z_0)} = 0. \quad (42)$$

Note that when a firm and worker bargain, they take the expected outcome of a worker bargaining with other firms as given. Thus $U(z_0)$ does not itself depend directly on $\mathcal{B}(z_0)$. Therefore, equation (42) implicitly characterizes a particular choice for $\mathcal{B}(z_0)$ from the Nash bargain.

A.4 Endogenous Separations and Limited Worker Commitment

This section introduces efficient endogenous separations and limited worker commitment into the baseline environment. To economize, we only discuss the parts of the model that change due to efficient separations or limited worker commitment; otherwise, the model is the same as the flexible incentive pay economy of the main text.

A.4.1 Economic Environment

Labor Market As in the baseline model of the main text, a large measure of risk-neutral firms match with unemployed workers according to a frictional matching technology. Fluctuations are driven by aggregate productivity z_t , and there is free entry to vacancy posting at a constant flow cost κ , as in the main text.

At the end of period $t-1$ an endogenous fraction s_t of workers separate from employment and enter unemployment. The unemployed search for new jobs, so u_t evolves as

$$u_t = u_{t-1} + s_t(1 - u_{t-1}) - \phi(\theta_{t-1})u_{t-1}(1 - s_t). \quad (43)$$

Preferences and Consumption Workers' preferences are identical to the model of the main text.

Firms and Wage Setting The value of a firm of posting a vacancy at time 0 is

$$\Pi_0 = q(\theta_0)\mathbb{E} \left[\sum_{t=0}^{\infty} \beta^t \left(\prod_{j=1}^t (1 - s_j) \right) (f(z_t, \eta_t) - w_t) \right] - \kappa, \quad (44)$$

where $\mathbb{E}$ conditions on the firm's information set at time 0 prior to meeting a worker. A vacancy is filled with probability $q(\theta)$. If a firm meets a worker, its value is the expected present value of the difference between production and wage payments, discounted by the firm's discount factor β as well as separation risk. Here, $\prod_{j=1}^t (1 - s_j)$ is the endogenous probability that a match survives until period t , which cumulates the probability $1 - s_j$ that a match survives period j . We entertain two possibilities for wage setting.

Flexible Incentive Pay Economy As in the main text, the firm observes realizations of both z_t and η_t , but does not observe worker's effort. When a firm and a worker meet, the firm offers the worker a contract to incentivize effort and maximize firm value. The innovation of this section is that the firm now has the additional option to vary the probability that the match separates in each date and state. For instance, if the expected present value of profits has turned negative, the firm may choose to terminate the contract. Thus the contract may be summarized by functions $w_t(\eta^t, z^t) \in [\underline{w}, \bar{w}]$, $a_t(\eta^{t-1}, z^t) \in [\underline{a}, \bar{a}]$ and a separation probability $s_t(\eta^t, z^t) \in [0, 1]$ for all t and all realizations of η^t and z^t . Let $(\mathbf{w}, \mathbf{a}, \mathbf{s})$ denote a contract, with $\mathbf{w} \equiv \{w_t(\eta^t, z^t)\}_{t=0, \eta^t, z^t}^\infty$, $\mathbf{a} \equiv \{a_t(\eta^{t-1}, z^t)\}_{t=0, \eta^{t-1}, z^t}^\infty$ and $\mathbf{s} \equiv \{s_t(\eta^t, z^t)\}_{t=0, \eta^t, z^t}^\infty$. Let $\mathcal{X}$ denote the space of possible contracts.

Value of a Filled Vacancy. Under the contract $(\mathbf{w}, \mathbf{a}, \mathbf{s})$, and initial productivity z_0 , the firm's expected present value of profits from posting a vacancy is

$$V(\mathbf{w}, \mathbf{a}, \mathbf{s}; z_0) = \sum_{t=0}^{\infty} \int \int \beta^t \mathcal{S}_t(\eta^t, z^t) (f(z_t, \eta_t) - w_t(\eta^t, z^t)) \tilde{\pi}_t(\eta^t, z^t | z_0, \mathbf{a}) d\eta^t dz^t, \quad (45)$$

where $\mathcal{S}_t(\eta^t, z^t) \equiv \prod_{j=1}^t (1 - s_{t-j}(\eta^{t-j}, z^{t-j}))$ is the probability that a match survives the sequence η^t, z^t ; and $\tilde{\pi}_t(\eta^t, z^t | z_0, \mathbf{a})$ is the probability of observing a realization of η^t and z^t given the initial z_0 and the contracted effort function $\mathbf{a}$, as in the main text. The risk-neutral firm discounts period t profits by the economy-wide discount rate β^t and the probability $\mathcal{S}_t(\eta^t, z^t)$ that the match survives t periods.

The contract maximizes the value of a filled vacancy

$$J(z_0) = \max_{\{w_t(\eta^t, z^t), a_t(\eta^{t-1}, z^t), s_t(\eta^t, z^t)\}_{t=0, \eta^t, z^t}^\infty \in \mathcal{X}} V(\mathbf{w}, \mathbf{a}, \mathbf{s}; z_0) \quad (46)$$

subject to the incentive compatibility and participation constraints described below, as well as a new set of constraints that captures limited commitment by the worker.

Incentive Constraints. The incentive compatibility condition is similar to the main text but now accounts for endogenous separation risk

$$\begin{aligned} [\text{IC}] : \mathbf{a} \in \arg\max_{\{\tilde{a}_t(\eta^{t-1}, z^t)\}_{t=0, \eta^{t-1}, z^t}^\infty} & \sum_{t=0}^{\infty} \left[\int \int \beta^t \mathcal{S}_t(\eta^t, z^t) \left[u\left(w_t(\eta^t, z^t), \tilde{a}_t(\eta^{t-1}, z^t)\right) - \Psi(s(\eta^t, z^t)) \right. \right. \\ & \left. \left. + \beta s(\eta^t, z^t) \int U(z_{t+1}) \pi(z_{t+1} | z_t) dz^{t+1} \right] \tilde{\pi}_t(\eta^t, z^t | z_0, \tilde{\mathbf{a}}) d\eta^t dz^t, \right. \end{aligned} \quad (47)$$

where $\Psi(s_j)$ represents a convex utility cost to the worker of searching for a new job.

Participation Constraint. Likewise, the participation constraint must also account for separation risk and becomes

$$[\mathbf{PC}] : \sum_{t=0}^{\infty} \left[\int \int \beta^t \mathcal{S}_t(\eta^t, z^t) \left[u(w_t(\eta^t, z^t), \tilde{a}_t(\eta^{t-1}, z^t)) - \Psi(s(\eta^t, z^t)) \right. \right. \\ \left. \left. + \beta s(\eta^t, z^t) \int U(z_{t+1}) \pi(z_{t+1}|z_t) dz_{t+1} \right] \tilde{\pi}_t(\eta^t, z^t | z_0, \tilde{\mathbf{a}}) d\eta^t dz^t \right] \geq \mathcal{E}(z_0). \quad (48)$$

Limited Commitment. Limited commitment and endogenous separations mean that after any history η^τ, z^τ the worker must rather stay in the match than separate, leading to a constraint that for each η^τ, z^τ :

$$[\mathbf{ES}] : \sum_{t=\tau}^{\infty} \mathbb{E} \left[\beta^{t-\tau} \mathcal{S}_t^\tau(\eta^t, z^t) \left[u(w_t(\eta^t, z^t), a_t(\eta^{t-1}, z^t)) - \Psi(s(\eta^t, z^t)) \right. \right. \\ \left. \left. + \beta s(\eta^t, z^t) \mathbb{E}[U(z_{t+1}) | z_t] \right] \middle| \eta^\tau, z^\tau \right] \geq U(z_\tau), \quad (49)$$

where $\mathcal{S}_t^\tau$ is the survival probability after time τ , $\mathcal{S}_t^\tau(\eta^t, z^t) \equiv \prod_{j=\tau+1}^t (1 - s_{t+\tau+1-j}(\eta^{t+\tau+1-j}, z^{t+\tau+1-j}))$.

Bargaining and ex ante utility. To close the flexible incentive pay economy, we again assume ex ante utility $\mathcal{E}(z_0)$ is given by a reduced-form “bargaining schedule” $\mathcal{B}(z_0)$ as in Section A.3.

Rigid Wage Economy The rigid wage economy is identical to the rigid wage economy of the main text, including the assumption of an exogenous separation rate s .

Equilibrium Given initial unemployment u_0 and a stochastic process $\{z_t, \eta_t\}_{t=0}^\infty$, an equilibrium is a collection of stochastic processes $\{\theta_t, u_t\}_{t=0}^\infty$ and functions $J(z), U(z), \mathcal{E}(z)$, and $(\mathbf{w}, \mathbf{a}, \mathbf{s})$ such that for all firms: (i) θ_t satisfies the free entry condition so that Π_t , given in equation (44), is equal to 0 for all t ; (ii) u_t satisfies the law of motion for unemployment (43); (iii) wage, effort, and separation functions $(\mathbf{w}, \mathbf{a}, \mathbf{s})$ satisfy the flexible incentive pay economy equations (46)-(49), or $w_t = \bar{w}$, $a_t = \bar{a}$ and $s_t = s$ in the rigid wage economy; (iv) the value of unemployment $U(z)$ is defined in the same way as the main text; (v) the value of employment is defined the same way as the main text for the in the rigid wage economy, or $\mathcal{E}(z) = \mathcal{B}(z)$ in the flexible incentive pay economy; and (vi) the value of a filled vacancy $J(z)$ is given by (46) in the flexible incentive pay economy or the same way as the main text for the rigid wage economy.

A.4.2 Equivalence of Rigid and Incentive Pay with Endogenous Separations

This subsection shows that, without bargaining power or fluctuations in outside options, the first-order response of market tightness is the same in the rigid wage economy and the flexible incentive pay economy with endogenous separations. For simplicity, we make the same assumptions as the main text, such as studying impulse responses in a neighborhood of the non-stochastic steady state.

Proposition 5. *Assume that the set of feasible contracts that satisfies the incentive constraints (47) and the participation constraint (48) is non-empty and compact. Also, assume that the production function is homogeneous of degree one in aggregate productivity z , z_t is a driftless random walk, and the optimal incentive contract at the non-stochastic steady state is unique. Finally, assume that the firm makes take it or leave it offers to workers, the flow value of unemployment is constant, and the optimal contract is unique. Then, the impact elasticity of market tightness to shocks to z_0 is*

$$\frac{d \ln \theta_0}{d \ln z_0} = \frac{1}{\bar{\nu}} \frac{1}{1 - \Lambda} \quad (50)$$

where Λ is the steady state labor share defined as

$$\Lambda \equiv \frac{\sum_{t=0}^{\infty} \mathbb{E} \beta^t \prod_{j=1}^t (1 - s_j^*) w_t^*}{\sum_{t=0}^{\infty} \mathbb{E} \beta^t \prod_{j=1}^t (1 - s_j^*) f(\bar{z}, \eta_t)}$$

where s_j^* and w_t^* denote choices of separations and wages along the optimal contract, where the expectation $\mathbb{E}$ is evaluated along the optimal contract, and $\bar{z}$ is the value of z_t at the aggregate steady state.

This theorem shows that the flexible incentive pay economy with endogenous separations has an equivalent response of tightness on impact to the rigid wage economy of the main text. Note that the dynamics of the rigid wage economy are still given by equation (15). Therefore, incentive wage cyclicalities do not affect the impact response of tightness with endogenous separations as long as the flexible incentive pay economy and the rigid wage economy are calibrated to the same steady state labor share. In the incentive pay economy with endogenous separations, the labor share depends on the optimal choice of separation rates, as well as the factors from the model of the main text, such as wages and effort.

We stress that this result leads to equivalence for impact elasticities, as Pissarides (2009) discusses. In general, the response of tightness to labor productivity shocks after impact

will be different in the rigid wage and flexible incentive pay economies because endogenous separations lead to additional dynamics of unemployment after the impact of the shock.

Intuitively, in the model with efficient endogenous separations, separations are an additional choice which the firm can optimize over. However, changes in the optimal separation choice after TFP shocks have no first-order effect on profits—just as neither changes in optimally chosen effort nor wages affect profits. Likewise, the optimal contract ensures that workers do not wish to leave the match. Reoptimizations by the worker as aggregate conditions change do not affect profits. This logic is again due to the envelope theorem.

A.4.3 Proof of Proposition 5

The free entry condition in the flexible incentive pay economy is

$$\frac{\kappa}{q(\theta)} = J(z_0),$$

where $J(z_0)$ is defined in equation (46). Taking derivatives and rearranging implies

$$\begin{aligned} \frac{d \ln \theta_0}{d \ln z_0} &= \frac{1}{\nu_0} \frac{d \ln J(z_0)}{d \ln z_0} \\ &= \frac{1}{\nu_0} \frac{z_0}{J(z_0)} \frac{dJ(z_0)}{dz_0}. \end{aligned} \tag{51}$$

With Ψ convex, the optimal separation rates s_j^* will be interior. Under the assumptions of the proposition, z_0 does not enter either the incentive constraints, the participation constraint, or the limited commitment constraints directly. Therefore we have

$$\begin{aligned} \frac{dJ(z_0)}{dz_0} &= \frac{\partial J(z_0)}{\partial z_0} = \frac{\partial}{\partial z_0} \sum_{t=0}^{\infty} \mathbb{E} \beta^t \prod_{j=1}^t (1 - s_j^*) (f(z_t, \eta_t) - w_t) \\ &= \sum_{t=0}^{\infty} \mathbb{E} \beta^t \prod_{j=1}^t (1 - s_j^*) (f_z(z_t, \eta_t)), \end{aligned} \tag{52}$$

where the first equality invokes the envelope theorem, using the same argument as Appendix Section A.1.1 and also using our assumption of a unique optimal contract in order to dispense with a sup operator; the second equality rewrites the definition of profits from equation (45) using the notation from the theorem and exploits that terms involving the participation, incentive, or limited commitment constraints vanish because z_0 does not enter them directly; and the final equality uses that z_t is a random walk.

Substituting in equations (51) and (52) implies

$$\begin{aligned}
\frac{d \ln \theta_0}{d \ln z_0} &= \frac{1}{\nu_0} \frac{z_0 \sum_{t=0}^{\infty} \mathbb{E} \beta^t \Pi_{j=1}^t (1 - s_j^*) (f_z(z_t, \eta_t))}{\mathbb{E} [\sum_{t=0}^{\infty} \beta^t \Pi_{j=1}^t (1 - s_j^*) (f(z_t, \eta_t) - w_t^*)]} \\
&= \frac{1}{\nu_0} \frac{\bar{z} \sum_{t=0}^{\infty} \mathbb{E} \beta^t \Pi_{j=1}^t (1 - s_j^*) (f_z(\bar{z}, \eta_t))}{\mathbb{E} [\sum_{t=0}^{\infty} \beta^t \Pi_{j=1}^t (1 - s_j^*) (f(\bar{z}, \eta_t) - w_t^*)]} \\
&= \frac{1}{\nu_0} \frac{\sum_{t=0}^{\infty} \mathbb{E} \beta^t \Pi_{j=1}^t (1 - s_j^*) (f_z(\bar{z}, \eta_t))}{\mathbb{E} [\sum_{t=0}^{\infty} \beta^t \Pi_{j=1}^t (1 - s_j^*) (f(\bar{z}, \eta_t) - w_t^*)]} \\
&= \frac{1}{\nu_0} \frac{1}{1 - \frac{\mathbb{E} [\sum_{t=0}^{\infty} \beta^t \Pi_{j=1}^t (1 - s_j^*) w_t^*]}{\mathbb{E} [\sum_{t=0}^{\infty} \beta^t \Pi_{j=1}^t (1 - s_j^*) (f(\bar{z}, \eta_t))]}},
\end{aligned}$$

where we use the assumption of an aggregate steady state in $\bar{z}$. □

B Numerical Appendix

B.1 Numerical Analysis: Calibration

Parameterizing the model. We parameterize the production function, utility function, ex ante utility, and information structure of the flexible incentive pay model of Section 2 following Edmans et al. (2012), allowing for cyclical ex ante utility as in section A.3.

Production function. The firm's production function is $y = z(a + \eta)$. Idiosyncratic profit shocks η are assumed to be i.i.d. over time and across individuals and normally distributed with zero mean and standard deviation σ_η . σ_η determines the extent to which firms can infer workers' effort, which is key for incentive pay.

Preferences. We assume that workers' utility function is given by $u(c, a) \equiv \ln c - \frac{a^{1+1/\epsilon}}{1+1/\epsilon}$. ϵ governs the Frisch elasticity of effort, which determines how costly effort is to workers.

Information structure. We make the "effort after noise" assumption as in Edmans et al. (2012): workers observe the idiosyncratic profit shock η before making an effort choice. Thus, there is an incentive compatibility constraint for each value of η . Following Edmans et al. (2012), we assume that a unique level of effort $a(z^t)$ is implemented regardless of the idiosyncratic shock η .¹⁹ However, effort varies with the history of aggregate productivity z^t .

Ex ante utility. We assume that firms make take-it-or-leave-it offers to workers who face procyclical unemployment benefits. Workers' flow unemployment benefits take the form $b(z) = \gamma z^\chi$. Here, γ specifies the level of unemployment benefits when $z = 1$, while χ deter-

¹⁹This assumption differs from the setup of the model of Section 2 and is for computational tractability. As Edmans et al. (2012) discuss, without this assumption, a closed form for the optimal contract is unavailable, which prohibits simulating the model.

mines the elasticity of unemployment benefits to aggregate productivity. This specification is a log-linear approximation of any differentiable $\mathcal{B}(z)$ function, including models in which workers and firms bargain over ex ante utility at the start of the contract. However, this specification is numerically tractable in that it abstracts from complications of bargaining and ensures that unemployed workers' value is given by the present discounted value of expected unemployment benefits. The parameter χ stands for all the reasons why the utility promised to workers at the beginning of an employment relationship may be cyclical—such as fluctuations in either the worker's outside option (changes in the value of unemployment) or her inside option (bargained utility)—and determines NWC.

We now characterize the behavior of wages under the optimal contract following Edmans et al. (2012), which will be useful for motivating our calibration strategy.

Proposition 6. *The earnings schedule in the optimal contract satisfies the following difference equation (given initial productivity z_0):*

$$\ln(w_t(\eta^t, z^t)) = \ln(w_{t-1}(\eta^{t-1}, z^{t-1})) + \psi h'(a_t)\eta_t - \frac{1}{2}(\psi h'(a_t)\sigma_\eta)^2, \quad (53)$$

where $\psi = 1 - \beta(1 - s)$ and $w_{-1}(z_0)$, which initializes the difference equation, is defined in the proof of the proposition.

The proof closely follows those of Edmans et al. (2012) and Doligalski et al. (2023).²⁰ Equation (53) characterizes wage growth. The pass-through of idiosyncratic shocks to wages, $\psi h'(a_t)\eta_t$, corresponds to incentives. If the marginal disutility of effort h' is high, there must be a high pass-through from η to wages to induce workers to supply the optimal effort level. To satisfy dynamic incentives, the pass-through of idiosyncratic productivity shocks to wages is scaled down by a quantity ψ that reflects discounting. Exponentiating equation (53), one observes that wages are a random walk: the expectation of wages in period $t + h$ is equal to the level of wages in period t . The random walk property is a consequence of the inverse Euler equation (Rogerson, 1985). Thus, rescaled wages at the start of the job, w_{-1}/ψ , are equal to the expected present discounted value (EPDV) of wage payments.

Calibration: Separating bargaining and outside options from incentives. Our goal is to infer the role of bargaining and outside options versus that of incentives in determining wage cyclicalities. We disentangle these forces with two sets of moments: the cyclicalities

²⁰We make three advances relative to Edmans et al. (2012): we introduce aggregate risk z_t , allow for pro-cyclical ex ante worker utility, and develop a global solution algorithm to efficiently simulate the model with labor market search.

of the wage for new hires, which informs non-incentive wages, and the pass-through of idiosyncratic firm output shocks into wages as well as the variance of workers' wage growth, both of which inform incentives. That is, wage fluctuations after the start of the match inform incentives, while wage fluctuations at the start of the contract inform NWC.

We calibrate the parameters of the labor search block largely following the standard practice of Petrosky-Nadeau and Zhang (2017).²¹ Productivity is assumed to follow an AR(1) process in logs, with autocorrelation parameter ρ_z , innovation $\zeta_t \sim \mathcal{N}(0, \sigma_z^2)$, and mean μ_z . We normalize μ_z such that $\mathbb{E}[z_t] = 1$. To account for the effects of effort fluctuations on labor productivity, we calibrate our monthly process for z such that the log of the quarterly average of z_t matches the autocorrelation and standard deviation of the quarterly log TFP series described in Fernald (2014), which accounts for variable capacity utilization in labor. We view the TFP series net of variable capacity utilization as a reasonable proxy for exogenous productivity, as labor utilization is a concept highly related to effort.²² This procedure implies a monthly autocorrelation $\rho_z = 0.966$ and standard deviation of shocks $\sigma_z = 0.0056$.²³

This leaves four parameters to internally calibrate: the variance of noise σ_η , the level and cyclical of ex ante utility χ and γ , and the effort elasticity ϵ . We target the variance of incumbent wage growth, the pass-through of firm shocks into wages, the cyclical of new hire wages, and the average unemployment rate. While we estimate all parameters jointly, these moments have intuitive mappings to particular parameters, which we explore below.

First, the variance of wage growth naturally informs the variance of idiosyncratic profit shocks σ_η . To see this, note that rearranging equation (53) shows that the monthly wage growth of job-stayers is given by $\Delta \ln w_t = \psi h'(a_t) \eta_t - 1/2 (\psi h'(a_t) \sigma_\eta)^2$. At an aggregate nonstochastic steady state, $a_t = a^{SS}$, for example, the cross-sectional variance of wage growth is given by $Var(\Delta \ln w) = \psi^2 h'(a^{SS})^2 \sigma_\eta^2$, which is closely tied to the value of σ_η . The firm provides *intertemporal incentives* by exposing the worker to wage-growth risk as in Sannikov (2008). We target a standard deviation of year-over-year wage growth of job-stayers of 0.064 as measured by Grigsby et al. (2021), where we calculate year-over-year wage growth in the model with stochastic z_t by iterating on equation (53) for job-stayers.²⁴

Second, the pass-through of firm-specific shocks to wages is informative of whether incentives are high powered within the contract, as in classic theories of moral hazard. In

²¹These parameters are the discount rate, the vacancy creation cost, the matching function, and the separation rate. We discuss the details in Appendix Section B.2.

²²Basu and Kimball (1997) find that variable capacity utilization explains approximately 40–60% of fluctuations in unadjusted TFP and that capacity utilization is procyclical.

²³We HP-filter the TFP data and model-simulated series with a smoothing parameter of $\lambda = 10^5$, following Shimer (2005), which removes a very low-frequency trend.

²⁴Hours are observable and thus contractible. We therefore consider earnings per hour—including base pay, bonuses, and overtime—to be the correct empirical counterpart of w_t .

particular, this pass-through helps us identify the parameter governing the disutility of effort ϵ . In our model, the expected pass-through from idiosyncratic output shocks to the wages of job-stayers is given by $\mathbb{E}[\partial \ln w / \partial \ln y] = \mathbb{E}[\psi h'(a)(a + \eta)]$, which is directly affected by $h'(a)$. The firm provides *intratemporal incentives* with the pass-through of output to wages. Intuitively, if $h'(a)$ is high, then workers would prefer not to supply more effort, so the firm must make wages highly dependent on output to incentivize effort.

A large literature seeks to estimate the pass-through to job-stayers' wages of firm-specific profitability shocks; Card et al. (2018) provides a comprehensive survey. We target an average pass-through of firm-level output shocks to wages of 0.039, the value estimated in Martins (2009), which is on the low end of the range reported by Card et al. (2018). Our targeting of a low pass-through value is likely to be a conservative choice, as low pass-through suggests that incentives are not high powered and therefore are a relatively unimportant determinant of wage variation.

Third, we identify γ , which pins down the level of unemployment benefits from the stochastic mean of unemployment. Average unemployment is determined by workers' job finding rates, which in turn are determined by expected profits per worker. γ directly influences expected profits because it governs workers' value of unemployment and shifts the level of the required wage payments to workers. We target an average unemployment rate of 6%, consistent with average U.S. unemployment between 1951 and 2019.

Fourth, we target the cyclicalities of new hire wages to inform the cyclicalities of nonemployment benefits χ . Conditional on the parameters governing incentives, the cyclicalities of new hire wages is highly informative of χ . Intuitively, if the worker's outside option is highly procyclical, so too is her promised utility, and thus, so too will be the present value of her wages. Since wages are a random walk in the optimal contract by equation (53), the cyclicalities of new hire wages strongly informs the cyclicalities of the present value of wages. We target a semielasticity of new hire wages to unemployment of -1 , which is at the high end of the range found by Grigsby et al. (2021) and Hazell and Taska (2022), and explore the robustness of our findings to this choice.

Our model links ex post wage pass-through to incentives and not to Nash bargaining. In the face of this particular concern, we target a conservative value of pass-through. Moreover, there is empirical evidence that pass-through is procyclical (Chan et al., 2023), which is consistent with our model and inconsistent with pass-through representing Nash bargaining.

Numerical results. Table B1 summarizes our calibration, while Table B2 examines the implications for various moments. We estimate that the elasticity of the disutility of effort ϵ is equal to 2.7. Note that standard estimates of micro labor supply elasticities, such as those

Table B1: Calibrated parameter values

Parameter	Description	Value	Source/Target
<i>Externally Calibrated</i>			
β	Discount Rate	0.990 ^{1/3}	Petrosky-Nadeau and Zhang (2017)
κ	Vacancy Creation Cost	0.450	Petrosky-Nadeau and Zhang (2017)
s	Separation Rate	0.031	CPS E-U Flow Rate
ρ	Autocorrelation: Agg. Productivity	0.966	Autocorrelation: Fernald (2014) TFP
σ_z	Cond. S.D. of Agg. Productivity	0.006	Uncond. S.D.: Fernald (2014) TFP
<i>Internally Calibrated</i>			
γ	Level: Unemployment Benefits	0.461	Average Unemployment Rate
ϵ	Elasticity: Disutility of Effort	2.713	Pass-Through: Profits to Wages
σ_η	S.D.: Idiosyncratic Profit η	0.532	S.D.: Job-Stayer Log Wage Growth
χ	Cyclicalilty: Promised Utility to Worker	0.467	New Hire Wage Cyclicalilty

computed by Chetty (2012), consider how hours vary with wages. Since hours are observable and contractible by the firm, the lower elasticities of hours need not have any relationship with the elasticity of unobservable effort. Intuitively, one might expect the elasticity of effort to be larger than that of hours: while many jobs have a fixed number of hours over which the worker has little control, workers may be able to adjust unobserved effort more elastically.

We find the level of unemployment benefits γ to be 0.46. This value is between the value chosen by Shimer (2005) to match the replacement rate of unemployment benefits (0.4) and that in Hagedorn and Manovskii (2008) to match aggregate wage cyclicalilty (0.955).²⁵

We estimate the standard deviation of idiosyncratic profit shocks to be $\sigma_\eta = 0.53$, similar to estimates in other labor search calibrations with idiosyncratic shocks (e.g., Schaal, 2017). This, coupled with a sizable elasticity of effort, suggests that incentive provision is a relatively important consideration for the firm. We estimate the cyclicalilty of flow unemployment benefits χ to be 0.47, implying moderately procyclical promised utility to the worker.²⁶

Table B2 compares key moments in both the data (Column (1)) and calibrated model (Column (2)). The top panel reports the moments that we target in the estimation. The model is able to fit the targeted moments very well. Most notably, we match the cyclicalilty of new hire wages almost exactly and, if anything, underestimate the pass-through of firm shocks to wages, suggesting that our estimate of the importance of incentives for wage cyclicalilty is likely a lower bound on its true importance.

The bottom panel of the table shows that the model generates approximately half of the unconditional volatility of aggregate unemployment observed in the data, which is an

²⁵Note, however, that unemployed workers do not need to supply effort in this model, which increases the effective flow unemployment value.

²⁶Chodorow-Reich and Karabarbounis (2016) estimate $\chi \approx 0.8$; however, the value of unemployment in our model is different from theirs because workers supply effort and do not have access to financial assets.

Table B2: Model fit to data moments

Moment	Description	Data (1)	Model (2)
<i>Targeted</i>			
$d\mathbb{E}[\ln w_0]/du$	Cyclicalities of new hire wages	−1.000	−1.001
$\mathbb{E}[\partial \ln w_t / \partial \ln y_{it}]$	Within-job pass-through of idiosyncratic shock	0.039	0.036
$\text{std}(\Delta \ln w_t)$	std(ln wage growth for job-stayers)	0.064	0.064
$\bar{u}_t$	Mean unemployment	0.060	0.060
<i>Untargeted</i>			
std (ln u_t)	Volatility of unemployment (quarterly)	0.203	0.103
Incentive share	Share of wage cyclicalities due to incentives	–	0.457

appropriate figure because labor productivity is not the sole determinant of unemployment fluctuations (Pissarides, 2009). Therefore, even though our main focus is the impulse response of unemployment, our calibrated model does match unconditional unemployment fluctuations reasonably well. Matching the micro moments of wage adjustment therefore generates significant unemployment volatility, the reasons for which we will discuss shortly.

Incentive wage cyclicalities. Now, we discuss our key numerical result: the model suggests that a significant share of wage cyclicalities is due to incentives. As a result, unemployment responds strongly to business cycle shocks despite wages being relatively procyclical.

The model calibration reveals in the final row of Table B2 that approximately 46% of the total wage cyclicalities is due to incentives. This may seem large. Non-base compensation, which may be associated with incentives, is relatively small for most workers. However, what matters for wage cyclicalities is whether the *marginal* dollar of wages paid is due to incentives or bargaining and outside options. If, for instance, 2% of compensation is incentive pay in the steady state but only incentive pay is cut in response to output shocks, then the share of wage cyclicalities attributable to incentives is 100%. Further, base wages may embed some incentive components if workers can be promoted after good performance.

Because NWC is relatively small, the impulse response of unemployment to business cycle shocks is relatively large. Table 3 reports a number of additional features of our model calibrated in a variety of ways. Column (1) reproduces the baseline calibration as in Table B2. The impulse response of market tightness to business cycle shocks is in the second row. Market tightness responds greatly to exogenous productivity shocks: the elasticity of market tightness to aggregate productivity is 13.6. In turn, unemployment is also volatile.

This occurs despite total wages being quite procyclical. The elasticity of the present

Table B3: Model moments: Alternative calibrations

Moment	Model: Source of wage flexibility			
	(1) Incentives + Bargaining	(2) Incentives	(3) Bargaining	(4) Bargaining: $\partial \mathbb{E}[\ln w_0]/\partial u = -0.54$
$d\mathbb{E}[\ln w_0]/du$	-1.00	-0.62	-1.00	-0.54
$d \ln \theta_0 / d \ln z_0$	13.6	17.8	10.4	13.3
$\text{std}(\ln u_t)$	0.10	0.15	0.08	0.10
$\mathcal{W}_0/\mathcal{Y}_0$	0.96	0.96	0.96	0.96
$d \ln \mathcal{W}_0 / d \ln z_0$	0.44	0.39	0.31	0.24
$d \ln \mathcal{Y}_0 / d \ln z_0$	0.70	0.88	0.51	0.51
NWC share	0.54	0.00	1.00	1.00

Notes: New hire wage cyclicality is targeted, while the second set of moments is untargeted. Column (1) is our baseline model. Column (2) sets $\chi = 0$ and does not target the cyclicality of new hire wages. Columns (3) and (4) fix effort $a = 1$, set wages to be constant within the contract, and do not target the standard deviation of wage growth or the pass-through. Column (4) targets a cyclicality of new hire wages of -0.54. The standard deviation of log unemployment is computed at quarterly frequency. x_0 denotes the value of variable x , evaluated at $\ln z = \mu_z$. $\mathcal{W}$ and $\mathcal{Y}$ refer to the expected present value of wage payments and output, respectively. “NWC share” is the share of wage cyclicality not accounted for by incentives.

value of expected wage payments with respect to productivity is 0.44. However, as we have discussed in previous sections, the stabilizing effect on unemployment of procyclical wages is offset by the amplifying effect of effort and incentives. Because of incentives, the response of the present value of output, $\mathcal{Y}_0$, to TFP shocks is a relatively large value of 0.70. As a result, profit fluctuations—and thus market tightness and employment fluctuations—are large despite the procyclicality of wages. The model implies a labor share (defined as $\mathcal{W}_0/\mathcal{Y}_0$) of 0.96, in line with, for instance, Hall (2005).²⁷

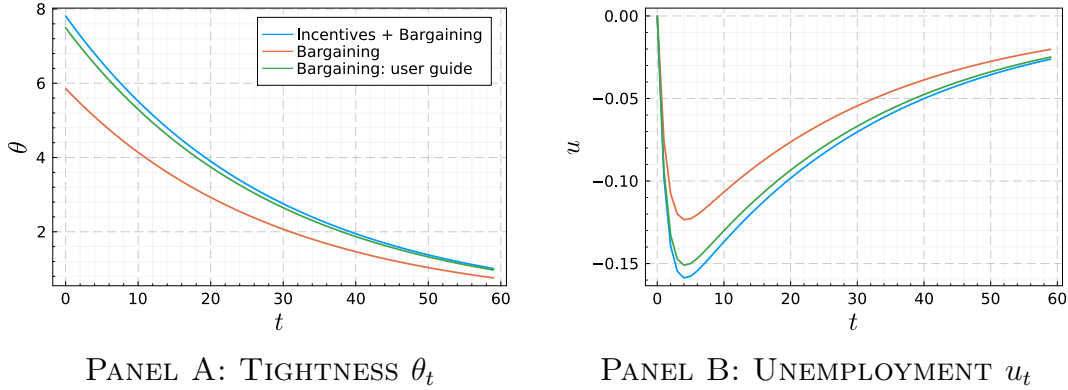
To emphasize the role of incentive wage cyclicality, we consider versions of our model that load all wage cyclicality in the data onto either incentives or bargaining and outside options. We present the calibration with only incentives in Column (2), which leads to a large impulse response of tightness in row 2.²⁸ Nevertheless, the incentives-only model still generates large wage cyclicality in row 1. This is a manifestation of our analytical results in a globally solved model. Column (3) presents a version of the model without incentives and with only bargaining, where the impulse response of tightness is relatively small, reflecting the dampening effect of NWC.²⁹

²⁷Since our model does not have capital, the labor share corresponds to the labor share of payroll and rents from search frictions in the labor market, excluding capital (Pissarides, 2000).

²⁸This calibration assumes that the cyclicality of ex ante utility is zero and do not target wage cyclicality.

²⁹This calibration turns off incentives by setting the variance of the idiosyncratic profitability shocks to $\sigma_\eta = 0$, exogenously fixing effort $a = 1$, setting $\epsilon = 1$, and setting wages to be fixed within a contract. We attribute all wage cyclicality in the data to the cyclicality of promised utility governed by χ .

Figure B3: Impulse response to shock to z_0 with bargaining-only and incentive pay models



Notes: The figure shows impulse responses for five years after a one-standard-deviation shock to z_0 . In Panel A, θ_t is shown in percentage deviations from the steady state (i.e., 100 times the log deviation). In Panel B, u_t is shown as deviations away from the steady state in percentage points (i.e., 100 times the deviation in levels). Further details on the construction of these impulse responses are described in Section B.2.4.

Calibrating simpler models. We argue that the simple version of our model in which all wage cyclicality is due to bargaining should target a new hire wage cyclicality given only by NWC—that is, a calibration in which wages are less procyclical than in the data. To illustrate the point numerically, we recalibrate the bargaining-only version of the model targeting a new hire wage cyclicality of -0.54, which is what we previously inferred to be non-incentive wage cyclicality.³⁰ Column (4) of Table B3 presents the results of this exercise.

The numerical results show that to produce the correct impulse response of market tightness in the simple model with only bargaining, calibrating to target NWC is crucial. When calibrated to NWC, the bargaining-only model features an elasticity of market tightness to exogenous shocks that is nearly identical (13.3) to that in the full model (13.6). Furthermore, both models generate an unconditional standard deviation of log unemployment rates of 0.10. The similar dynamics arise because the two models imply similar ex ante utility cyclicality even though overall wage flexibility is different: the simple bargaining-only model of column (4) estimates an elasticity of unemployment benefits $\chi = 0.47$, nearly identical to that found under the full model.

Figure B3 plots the impulse of market tightness (Panel A) and unemployment (Panel B) in response to a one-standard-deviation increase in aggregate productivity z , which decays according to an AR(1). The blue line is the response in the full model with both incentives and bargaining. The red line is the response in the bargaining-only model calibrated to the full wage cyclicality in the data. The green line is the response in the bargaining-only model calibrated to our estimate of NWC in the data. The response of both market tightness and

³⁰We normalize $\epsilon = 1$ for this exercise and solve for fixed wages within the contract. We also drop the standard deviation of log wage growth and the average pass-through of firm shocks as targeted moments.

unemployment is approximately 25% less pronounced in the bargaining-only model than in the full model with incentives and bargaining. However, the impulse responses of both tightness and unemployment are nearly identical in the full model and the bargaining-only model calibrated to relatively rigid wages.

Robustness. The key numerical result of this section is that a significant share of wage cyclicalities is due to incentives, leading to volatile unemployment dynamics despite wages being relatively procyclical. Appendix B.2.5 probes the robustness of this result. Tables B4 and B5 report the estimated parameters and model-implied moments, respectively, when we target different values of wage cyclicalities ranging from -0.5 to -1.5 . We find that the share of wage cyclicalities attributable to incentives declines as we increase the target cyclicalities of new hire wages. However, the elasticity of incentive wages to unemployment is always large and relatively stable between -0.37 and -0.49 .

To account for uncertainty in our wage pass-through target, Appendix Figure B6 reports the estimate of the incentive wage cyclicalities share as one varies the elasticity of effort supply ϵ , recalibrating the rest of the parameters. The estimated share of wage cyclicalities due to incentives is increasing in ϵ , rising to 52% for $\epsilon = 5$ and falling to 23% for $\epsilon = 0.5$.

Next, we study the robustness with respect to our TFP shock series. As noted previously, incentives lead to changes in measured productivity through endogenous effort fluctuations. Our utilization-adjusted TFP series imperfectly corrects for these effort changes. Therefore, we also internally calibrate the exogenous productivity process in our incentive pay model to match moments of average labor productivity in the data. Appendix Tables B4 and B6 report the estimated parameters and model-implied moments, respectively. Calibrated thus, the model continues to infer that incentives account for about 40% of overall wage cyclicalities and a large response of market tightness to productivity shocks.³¹

Taking stock, we find that a relatively large share of wage cyclicalities in the data is attributable to incentives despite our conservative calibration. Therefore, our model generates a large impulse response of unemployment despite the cyclicalities of wages.

B.2 Numerical Details

We calibrate the model such that t represents a month. Specifically, we set the discount rate β to $0.99^{1/3}$, the vacancy creation cost to 0.45 and employ a matching function given by $m(u, v) = uv(u^\iota + v^\iota)^{-1/\iota}$ so that $q(\theta) = (1 + \theta^\iota)^{1/\iota}$, which is bounded between 0 and

³¹In Appendix Tables B4 and B6, we also recalibrate the bargaining-only model to target average labor productivity. The bargaining-only model continues to have a significantly smaller impulse response of tightness than does the full model and requires exogenous productivity shocks to be approximately twice as volatile as in the full model to match output fluctuations.

1. We set $\iota = 0.9$ by nonlinear least squares to match the empirical relationship between aggregate market tightness and job-finding rates. We set the exogenous separation rate $s = 0.031$ to the average monthly separation rate in the Current Population Survey (CPS) from 1951 to 2019. This implies that the pass-through parameter ψ equals 0.034. Separation rates and job-finding rates are both adjusted for time aggregation following Shimer (2005). We measure empirical labor market tightness as job openings from the Job Openings and Labor Turnover Survey (JOLTS) divided by household unemployment in the CPS. Our labor market tightness series spans from 2001 to 2019 (JOLTS begins in December 2000).

We discretize the AR(1) productivity process for $\ln z_t$ onto a finite grid: $z \in \mathcal{Z} = [\underline{z}, \dots, \bar{z}]$ following Rouwenhorst (1995). We set the number of gridpoints to 13.

We now rewrite the key equations in our numerical model recursively, given the Markovian structure for productivity. Let $\pi(z'|z)$ denote the probability of aggregate productivity transitioning from z to z' . Recall that the optimal effort schedule, given an initial z_0 and current z , satisfies

$$a(z; z_0) = \left[\frac{za(z; z_0)}{\psi \left(Y(z_0) - \frac{\kappa}{q(\theta(z_0))} \right)} - \frac{\psi}{\epsilon} (h'(a(z; z_0)) \sigma_\eta)^2 \right]^{\frac{\epsilon}{1+\epsilon}}.$$

Let $\tilde{Y}(z; z_0)$ denote the EPDV of future output, conditional on effort $a(\cdot; z_0)$ and current productivity z , given by

$$\tilde{Y}(z; z_0) = za(z; z_0) + \sum_{z' \in \mathcal{Z}} \beta (1-s) \tilde{Y}(z'; z_0) \pi(z'|z).$$

It follows that $Y(z_0) = \tilde{Y}(z_0; z_0)$. Note that the optimal effort depends on z_0 through $Y(z_0)$ and $\theta(z_0)$, which are both equilibrium objects in our model. Define the worker's expected present discounted utility from starting work at z_0 , $\tilde{\mathcal{E}}(z_0)$, taking as given the effort schedule $a(\cdot; z_0)$ and the wage schedule $w(\cdot; z_0)$ defined in Proposition 6:

$$\tilde{\mathcal{E}}(z_0) = \frac{1}{\psi} \ln w_{-1}(z_0) + \mathbb{E}_z \left[- \sum_{t=0}^{\infty} [\beta(1-s)]^t \frac{1}{\psi} \frac{1}{2} (\psi h'(a(z_t; z_0)) \sigma_\eta)^2 - \sum_{t=0}^{\infty} [\beta(1-s)]^t h(a(z_t; z_0)) + \sum_{t=0}^{\infty} [\beta(1-s)]^t \beta s \omega(z_{t+1}) \right],$$

where

$$\omega(z) = \mathbb{E} \left[\sum_{t=0}^{\infty} \beta^t \ln b(z_t) \mid z_0 = z \right].$$

It is helpful to re-define the term in brackets in the above expression as $W(z_0; z_0)$, where

$$W(z; z_0) = -\frac{1}{\psi} \frac{1}{2} (\psi h'(a(z; z_0)) \sigma_\eta)^2 - h(a(z; z_0)) + \sum_{z' \in \mathcal{Z}} \beta s \omega(z') \pi(z'|z_0) + \sum_{z' \in \mathcal{Z}} \beta (1-s) W(z'; z_0) \pi(z'|z_0).$$

Finally, we define an implicit, auxiliary function for effort $\tilde{a}$ with arguments z , $\tilde{Y}$, and $\tilde{q}$ (subsuming any dependence on z_0) that is useful when solving the model numerically:

$$\tilde{a}(z, \tilde{Y}, \tilde{q}) = \left[\frac{z\tilde{a}}{\psi(\tilde{Y} - \kappa/\tilde{q})} - \frac{\psi}{\epsilon} (h'(\tilde{a}) \sigma_\eta)^2 \right]^{\frac{\epsilon}{1+\epsilon}}.^{32}$$

B.2.1 Algorithm to solve for the optimal contract, given z_0

Fix an initial $z_0 \in \mathcal{Z}$. To solve for the optimal contract beginning at z_0 , we perform a bisection search over the job-filling rate $q(z_0)$. Let n index iterations over our guess of $q(z_0)$. Then, for a given $q^n(z_0)$, we solve for the optimal effort schedule $\tilde{a}^n(\cdot)$ and the EPDV of output $Y^n(z_0)$ as a fixed point problem. With values of $Y^n(z_0)$ and $q^n(z_0)$, we can construct $w_1^n(z_0)$, the initialization for the difference equation governing the wage schedule, and recursively solve for the EPDV of the utility offered by the contract $\mathcal{E}^n(z_0)$. We then check whether $\mathcal{E}^n(z_0) = \omega(z_0)$, as implied by TIOLI offers, and accordingly update the lower and upper bounds for the next iteration, $\underline{q}^{n+1}$ and $\bar{q}^{n+1}$, respectively. We continue this process until convergence of $q(z_0)$. Below, we describe the algorithm in further detail.

1. Set $n = 1$. Set $\underline{q}^n = 0$, and $\bar{q}^n = 1$.
2. Set $q^n(z_0) = \frac{1}{2}(\underline{q}^n + \bar{q}^n)$.
3. Set $k = 1$. Make initial guess for $Y^{k,n}(z|z_0)$ for $z \in \mathcal{Z}$.
4. Update $Y^{k+1,n}(\cdot; z_0)$ as

$$Y^{k+1,n}(z; z_0) = z\tilde{a}(z, Y^{k,n}(z; z_0), q^n(z_0)) + \sum_{z' \in \mathcal{Z}} \beta (1-s) Y^{k,n}(z'; z_0) \pi(z'|z)$$

5. Repeat (4) until $\|Y^{k,n+1}(\cdot; z_0) - Y^{k,n}(\cdot; z_0)\| < \delta_1$ for some small tolerance $\delta_1 > 0$. Define the object $Y^n(z_0) = Y^{k,n}(z_0; z_0)$. Define $\tilde{a}^n(z) = \tilde{a}(z, Y^n(z_0), q^n(z_0))$.
6. Solve for $w_{-1}^n(z_0)$ using the free entry condition:

³²For general ϵ , we numerically solve for a_t using a root-finder, restricting attention to positive roots.

$$w_{-1}^n(z_0) = \psi \left(Y^n(z_0) - \frac{\kappa}{q^n(z_0)} \right).$$

7. Set $j = 1$. Make initial guess for $W^{j,n}(z_0; z)$.

8. Update $W^{j+1,n}(\cdot; z_0)$ as

$$W^{j+1,n}(z; z_0) = -\frac{1}{\psi} \frac{1}{2} (\psi h'(\tilde{a}^n(z)) \sigma_\eta)^2 - h(\tilde{a}^n(z)) + \sum_{z' \in \mathcal{Z}} \beta s \omega(z') \pi(z'|z) + \sum_{z' \in \mathcal{Z}} \beta (1-s) W^{j,n}(z'; z_0) \pi(z'|z)$$

9. Repeat (8) until $\|W^{j,n+1}(\cdot; z_0) - W^{j,n}(\cdot; z_0)\| < \delta_2$ for some small tolerance $\delta_2 > 0$.

Define $\mathcal{E}^n(z_0) = \frac{1}{\psi} \ln w_{-1}^n(z_0) + W^{j,n}(z_0; z_0)$.

10. If $\mathcal{E}^n(z_0) > \omega(z_0)$ then set $\bar{q}^{n+1} = q^n(z_0)$. If $\mathcal{E}^n(z_0) < \omega(z_0)$, then set $\underline{q}^{n+1} = q^n(z_0)$. Recall that with TIOLI offers, $\mathcal{E}(z_0) = \omega(z_0)$. Note that $\omega(z_0)$ can be computed by a simple value function iteration.

11. Repeat steps (2)-(10) until $|\mathcal{E}^n(z_0) - \omega(z_0)| < \delta_3$ for some small tolerance $\delta_3 > 0$ to obtain $q(z_0)$.

12. Define $\theta(z_0) = q^{-1}(q(z_0))$, where $q(\theta) = \frac{1}{(1+\theta^\epsilon)^{1/\epsilon}}$.

We repeat this procedure for all values of $z_0 \in \mathcal{Z}$ to obtain the equilibrium objects $Y(z_0)$, $w_{-1}(z_0)$, and $a(\cdot; z_0)$. It takes less than half of a second to solve for the optimal contract for a given z_0 with the parameters from our baseline calibration.

Our set of targeted moments includes two moments that depend on within-contract, idiosyncratic realizations: the standard deviation of annual (YoY) wage growth ($\text{std}(\Delta \ln w_{it})$) and the pass-through from idiosyncratic shocks to firm profits to wages ($\partial \ln w_{it} / \partial \ln y_{it}$), and two moments which can be computed from aggregate time series simulated in the model: the cyclicalities of new hire wages ($\partial \mathbb{E}[\ln w_0] / \partial u$) and average unemployment ($\bar{u}_t$). To compute these moments for a given set of parameters $\Omega := \{\epsilon, \sigma_\eta, \chi, \gamma\}$, we solve the model for each initial $z_0 \in \mathcal{Z}$ following the procedure outlined in Section B.2.1 to obtain $a(\cdot|z_0)$, $w_{-1}(z_0)$, and $\theta(z_0)$. We then simulate the economy with aggregate shocks and compute moments.

Simulating $\text{std}(\Delta \ln w_{it})$ and $\mathbb{E}[\partial \ln w_{it} / \partial \ln y_{it}]$. We simulate a panel of $I = 50,000$ idiosyncratic η_{it} shocks of length $T = 1,500$ (and one sequence of aggregate z_t shocks of length T). For each period t and worker i , we simulate separations and job-finding shocks consistent with the exogenous probability of separation s and endogenous job-finding probability

$\phi(\theta(z_t))$.³³ All workers are employed at the beginning of $t = 0$. During job spells and given realizations of z_t and η_{it} , we can compute log wages and the pass-through for each worker according to the equations derived in Section B.1. For job spells that last at least 13 months, we can compute YoY log wage growth as $\ln w_{i,t+12} - \ln w_{it}$ (for each year of employment). We discard the first $t_{\text{burn-in}} = 500$ periods as a burn-in period. We then compute the pooled variance of YoY log wage growth and the average monthly pass-through across all job spells/periods of employment for $t \geq t_{\text{burn-in}}$. Cross-sectional and longitudinal data on job spells/periods of employment (job-stayers) are interchangeable in this setting.

Simulating $d\mathbb{E}[\ln w_0]/du$ and $\bar{u}_t$. We simulate 10,000 z_t sequences of length $T = 828$ periods (with an additional burn-in period of length 500 periods), corresponding to monthly observations for the 1951-2019 period. For each z_t path, we can compute the path for unemployment as

$$u_{t+1} = u_t + s(1 - u_t) - \phi(\theta(z_t))u_t(1 - s)$$

given initial condition $u_0 = 0.06$. The expected log wage of new hire wages is

$$\mathbb{E}_{\eta_{it}}[\ln w_0(z_t)] = \ln w_{-1}(z_t) - \frac{1}{2}(\psi h'(a(z_t|z_t))\sigma_\eta)^2.$$

We compute $\bar{u}_t$ as the average unemployment u_t for $t \geq t_{\text{burn-in}}$. We measure $d\mathbb{E}[\ln w_0]/du$ in the model by running an OLS regression of $\mathbb{E}[\ln w_0](z_t)$ on u_t and a constant in the simulated data for $t \geq t_{\text{burn-in}}$. We report cross-simulation averages for both moments.

B.2.2 Estimation Algorithm

We implement the Tik-Tak algorithm, a multi-start global optimization algorithm, as described by Arnoud et al. (2019), to minimize the following objective function

$$J(\Omega) = (\tilde{m}(\Omega) - m)'W(m)(\tilde{m}(\Omega) - m),$$

where Ω is a vector of the parameters to be estimated, $\tilde{m}(\Omega)$ is a vector of the targeted moments computed using the model simulated data given the parameter vector Ω , and m is the vector of targeted empirical moments. The weight matrix W satisfies $W_{j,j} = |1/m_j|$ for each targeted moment j (and 0, otherwise). Thus, the objective function to minimize is the sum of squared percentage differences between simulated and empirical moments to account for differences in scale between the targeted moments. We have experimented with different derivative-free local optimization algorithms, such as BOBYQA and the Nelder-

³³This procedure includes composition effects of initial z_0 on the employment contracts.

Mead Simplex Algorithm, for the local optimization step. All estimation results reported in the paper correspond to solutions obtained using a combination of the Nelder-Mead Simplex Algorithm and BOBYQA algorithm with 1,000 initial points. We implement a pre-testing stage to detect promising regions of the parameter space by evaluating the objective function at 50,000 initial points drawn from Sobol sequences; we use the 1,000 points that yield the lowest values of the objective function as the initial points in the global search.

Technical detail on the participation constraint In some situations during the calibration, $q(z_0)$ may hit its upper bound of 1 with $\mathcal{E}(z_0) < \omega(z_0)$, violating the participation constraint. In this case, the implied job-finding rate is 0. Therefore, the value of unemployment $U(z_0)$ (before matching, at the beginning of the period) is equal to $\mathcal{B}(z_0)$. When $q(z_0) = 1$ and the participation constraint is violated, we can still simulate moments, but the implied new hire wage for z_0 would not be an observed wage, as $f(z_0) = 0$. The other moments would not be affected, given that we simulate employment spells and wage contracts in accordance with the endogenous job-finding probabilities.

We do not simulate moments when $\mathcal{E}(z_0) < \omega(z_0)$ binds for values of $\ln z_0$ within three unconditional standard deviations of μ_z . Instead, we penalize the parameters for which this occurs in a way that scales with the size of the deviation $|\mathcal{E}(z_0) - \omega(z_0)|$. We do not penalize violations for extreme z_0 as the probability of reaching extreme z_0 is low, and it may be reasonable to expect that the constraint $q(\theta(z_0)) \leq 1$ will bind for extremely low z_0 . This constraint is related to a binding nonnegative profit constraint, given that the zero profit condition is imposed within the algorithm to solve for the optimal contract via $w_{-1}(z_0)$. We have explored alternative approaches to handling participation constraint violations. In particular, the baseline results are largely unchanged when we penalize violations for $\ln z$ within five standard deviations of μ_z , which includes our entire discretized productivity grid.

B.2.3 Calculating Incentive Wage Cyclicity

Non-incentive wage cyclicity reflects movements in the promised utility of workers. For a given calibration, we calculate how the value of a filled job moves with exogenous productivity $dJ(z_0)/dz_0$. The “direct effect” of z_0 on the expected present discounted value of profits per worker, given the AR(1) process for $\ln z$, can be approximated as

$$\frac{dJ(z_0)}{dz_0} \stackrel{Direct}{=} \sum_{t=0}^{\infty} (\beta(1-s))^t \frac{\mathbb{E}_0[a^*(z_t)\rho^t z_t]}{z_0}.$$

That is, the direct effect is the effect that z has on profits holding fixed the optimally contracted choice of effort and wages. We calculate NWC as

$$\text{NWC}(z_0) = \frac{dJ(z_0)}{dz_0} - \frac{dJ(z_0)^{\text{Direct}}}{dz_0}.$$

The share of wage fluctuations attributable to incentives is then the negative of one minus $\text{NWC}(z_0)$ divided by the cyclicalty of the expected present discounted value of wage payments $dW^*(z_0)/dz_0$.

B.2.4 Construction of Impulse Responses

We compute the impulse response to a one (conditional) standard deviation (σ_z) shock to $\ln z_0$ in an economy that is at an aggregate non-stochastic steady state. In particular, we construct nonlinear perfect foresight impulse responses to a one-time shock to productivity at time 0 that decays at rate ρ_z . We define the non-stochastic steady state of $\log z$ to be 0, dropping the normalization of μ_z that ensures $\mathbb{E}[z_t] = 1$ given that $\mu_z \approx 0$.

We first solve for the non-stochastic steady state of the model, where $z_t = z_{ss} = 1$, $\theta_t = \theta(z_{ss})$, and $u_t = u_{ss} = \frac{s}{s + \phi(\theta(z_{ss}))(1-s)}$ for all t . We next solve for the path of $\theta_t(\{z_s\}_{s \geq t})$, given a sequence of shocks $\{z_t\}$. Finally, we solve for the path of unemployment u_t , given the path of θ_t , setting $u_0 = u_{ss}$.³⁴ We construct these impulse responses in a finite horizon contract setting and set the length of the contract, T , to be 240 model periods (20 years), which is a close approximation to the infinite horizon contract environment.³⁵

B.2.5 Additional Numerical Results

This section reports additional quantitative results for alternative calibrations. Table B4 reports estimated parameters for our robustness exercises. Table B5 reports moments when we target different values for the cyclicalty of new hire wages. Each column corresponds to a recalibration of the model. Similarly, Table B6 reports implied moments when we internally calibrate the process for exogenous labor productivity to match average labor productivity (ALP). ALP is the seasonally adjusted, quarterly average output per hour for all workers in the nonfarm business sector, as reported by the BLS. Figure B6 reports the estimated value

³⁴The calibration was done for the infinite horizon contract environment and targeted the stochastic mean of unemployment rather than steady state unemployment rate as implied in a non-stochastic model. Therefore, the steady state across the models need not be the same, although they are very close in practice.

³⁵There is an additional term in the law of motion for unemployment in the finite horizon contract setting because workers separate with probability one after they have completed their contract without experiencing a separation shock. However, the measure of workers that do not separate by time T is essentially zero, given that $T = 240$. Therefore, we ignore this inflow into unemployment in this numerical exercise.

of the incentive wage cyclical share for various imposed values of ϵ , allowing all other parameters to be recalibrated. The X on the plot reports our baseline estimate for ϵ .

Table B4: Alternative calibration strategies: Internally calibrated parameters

Parameter	$\partial \mathbb{E}[\ln w_0]/du$ target				Internal Calibration: ALP	
	-0.5	-0.75	-1.25	-1.5	Full	Bargaining Only
σ_η	0.530	0.533	0.533	0.533	0.528	0.000*
χ	0.203	0.341	0.549	0.609	0.465	0.617
γ	0.488	0.454	0.461	0.454	0.537	0.583
ϵ	2.047	2.949	2.744	2.956	1.377	1.000*
ρ_z	0.966*	0.966*	0.966*	0.966*	0.985	0.977
σ_z	0.006*	0.006*	0.006*	0.006*	0.002	0.005

Notes: Table reports estimated parameters for our alternative calibration strategies. The first four columns change the target of new hire wage cyclical share for our full model with both incentives and bargaining. The final two columns internally calibrate the exogenous productivity process to match moments of measured labor productivity under our full model and model with only bargaining. Asterisks indicate imposed parameters.

Table B5: Varying cyclical share of new hire wages: Simulated model moments

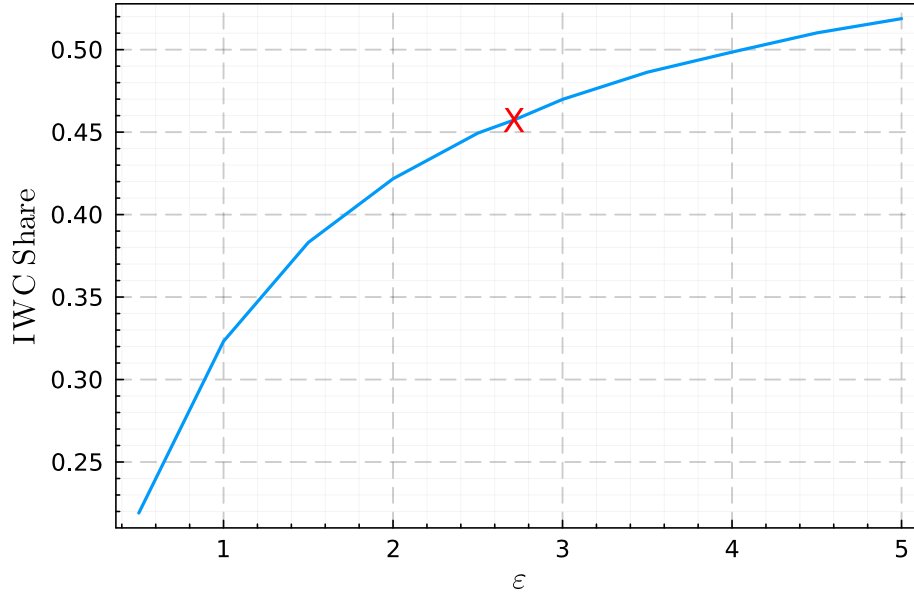
Moment	Model: $\partial \mathbb{E}[\ln w_0]/du$ target			
	-0.50	-0.75	-1.25	-1.50
$d\mathbb{E}[\ln w_0]/du$	-0.50	-0.75	-1.25	-1.50
$\text{std}(\ln u_t)$	0.16	0.13	0.09	0.08
$d \ln \theta_0 / d \ln z_0$	17.9	15.8	12.0	10.5
Incentive Wage Cyclical share	0.73	0.59	0.38	0.33
Incentive Wage Cyclical share	-0.37	-0.44	-0.47	-0.49

Notes: New hire wage cyclical share is targeted, while the second set of moments are untargeted. $\text{std}(\ln u_t)$ is the unconditional standard deviation of the log of the quarterly average of the monthly unemployment rate, HP-filtered with smoothing parameter $\lambda = 10^5$. x_0 denotes the value of variable x , evaluated at $\ln z = \mu_z$. Incentive Wage Cyclical share is the share of wage cyclical share that is due to incentives.

Table B6: Internally calibrating labor productivity process: simulated model moments

Moment	Data	Model: Source of wage flexibility	
		(1) Incentives + Bargaining	(2) Bargaining
ρ_y	0.89	0.89	0.89
σ_y	0.02	0.02	0.02
$\text{std}(\ln u_t)$	0.20	0.07	0.09
$d \ln \theta_0 / d \ln z_0$	-	18.7	11.6
$\mathcal{W}_0 / \mathcal{Y}_0$	-	0.96	0.96
$d \ln \mathcal{W}_0 / d \ln z_0$	-	0.55	0.37
$d \ln \mathcal{Y}_0 / d \ln z_0$	-	0.92	0.61
Incentive share	-	0.40	0.00

Notes: New hire wage cyclicality is targeted, while the second set of moments are untargeted. $\text{std}(\ln u_t)$ is the unconditional standard deviation of the log of the quarterly average of the monthly unemployment rate, HP-filtered with smoothing parameter $\lambda = 10^5$. x_0 denotes the value of variable x , evaluated at $\ln z = \mu_z$. ρ_y and σ_y are the autocorrelation and unconditional variance of measured average labor productivity.

Figure B6: Incentive wage cyclicality share for different calibrations of ϵ 

Notes: Figure reports the estimated share of wage cyclicality due to incentives at $\ln z = \mu_z$ as we vary the disutility of effort ϵ . To produce this figure, we first impose a value of ϵ and then recalibrate our model to match all four of our calibration targets.

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